

Networking Introduction

网络入门 · 中英对照译文

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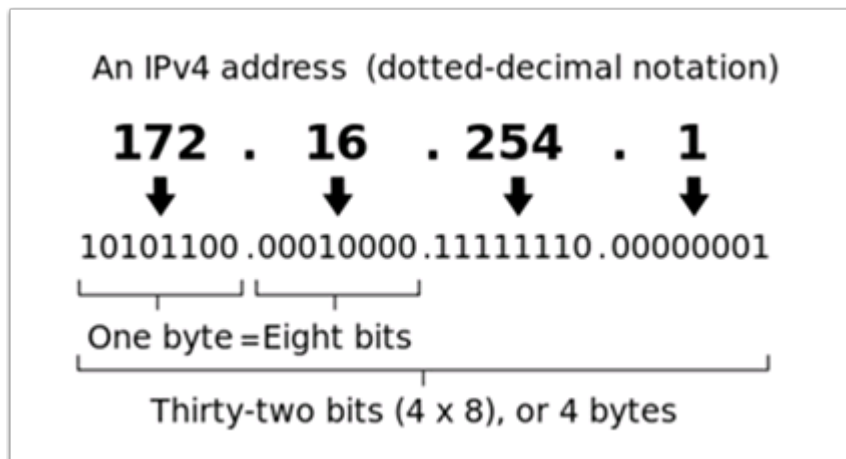
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Networking Basics / 网络基础

What is an IP Address? / 什么是 IP 地址?

An IP address is a unique string of numbers, separated by periods that identifies each device on a network. Each IP address is divided up into 4 sections (octets) ranging from 0-255.

IP 地址是一串用句点分隔的唯一数字，用于标识网络中的每台设备。每个 IP 地址分为 4 段（八位字节），每段的取值范围为 0-255。



The octet breakdown of an IP address and how it totals 4 bytes. / IP 地址的八位字节划分及其如何组成 4 字节。

As shown above, this means that each IP address is a 32-bit address meaning there are 2^{32} addresses, or nearly 4,300,000,000 addresses possible. However, most of these are used publicly for things like web servers.

如上图所示，每个 IP 地址是一个 32 位地址，这意味着可能的地址共有 2^{32} 个，即接近 4,300,000,000 个。不过，其中大多数已作为 Web 服务器等用途的公网地址使用。

This brings up our **first key point** of IP Addressing: Each device on the network must have a unique IP address. No two devices can have the same IP address, otherwise collisions will occur.

由此引出 IP 地址的**第一个要点**：网络中的每台设备都必须有唯一的 IP 地址。两台设备不能使用相同的 IP 地址，否则会发生冲突。

Since there are only 4 billion addresses, and there are more than 4 billion computers connected to the internet, we need to be as efficient as possible with giving out IP addresses. This brings us to public vs. private addresses.

由于地址只有 40 亿个，而连接到互联网的计算机已超过 40 亿台，我们必须尽可能高效地分配 IP 地址。这就引出了公网地址与私有地址的区别。

Public vs Private IP Addresses / 公网与私有 IP 地址

To be efficient with using IP Addresses, the idea of "Reserved IP Ranges" was implemented. In short, this means that there are ranges of IP Addresses that will never be assigned to web servers, and will only be used for local networks, such as those in your house.

为了高效使用 IP 地址，人们设置了“保留的 IP 地址范围”。简而言之，某些 IP 地址范围永远不会分配给 Web 服务器，只用于本地网络，例如家庭网络。

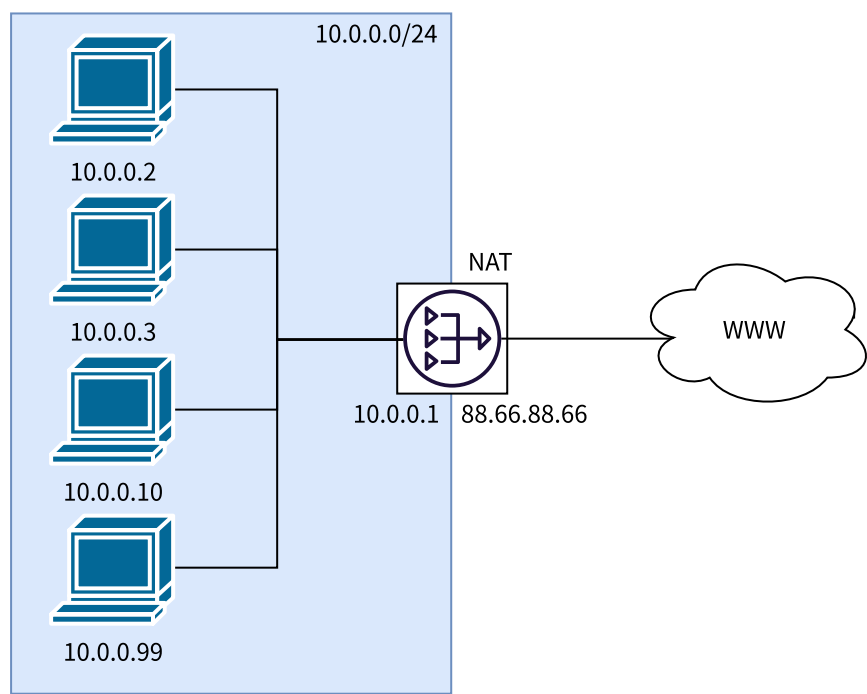
Key point #2: Unless you are directly connecting to your internet provider's basic modem (no router function), your device will have an IP Address in one of these ranges. This means that at any local network, such as: your school, work office, home, etc., your device will 99% of the time have an IP address in a range listed below:

要点 #2: 除非设备直接连接到互联网服务提供商的基础调制解调器（不具备路由功能），否则设备的 IP 地址会处于这些范围之一。这意味着在学校、办公室、家庭等本地网络中，设备的 IP 地址有 99% 的概率落在下表所列范围内：

Class / 类别	Bits / 位数	Start Address / 起始地址	End Address / 结束地址	Number of Addresses / 地址数量
A	24	10.0.0.0	10.255.255.255	16,777,216
B	20	172.16.0.0	172.31.255.255	1,048,576
C	16	192.168.0.0	192.168.255.255	65,536

These reserved ranges let us assign one "unreserved IP Address" to an entire house, and then use multiple addresses in a reserved range to connect more than one computer to the internet. A process on the home's internet router known as **NAT** (Network Address Translation), handles the process of keeping track which private IP is requesting data, using the public IP to request that data from the internet, and then passing the returned data back to the private IP that requested it. This allows us to use the same reserved IP addresses for many local networks, without causing any conflicts. An image of this process is presented below.

这些保留地址范围让我们可以给整户家庭分配一个“非保留 IP 地址”，再使用保留范围内的多个地址将多台计算机接入互联网。家庭互联网路由器上的 **NAT**（网络地址转换）负责跟踪是哪一个私有 IP 请求了数据，用公网 IP 向互联网请求数据，再将返回的数据转发给发起请求的私有 IP。这样，许多本地网络就可以重复使用相同的保留 IP 地址，而不会互相冲突。下图展示了这一过程。



Devices on the private network send their traffic through the NAT device to communicate to the outside network and vice versa. / 私有网络中的设备通过 NAT 设备与外部网络通信，外部网络也通过该设备与私有网络通信。

NOTE: For the FRC® networks, we will use the 10.0.0.0 range. This range allows us to use the 10.TE.AM.xx format for IP addresses, whereas using the Class B or C networks would only allow a subset of teams to follow the format.

说明: 对于 FRC® 网络, 我们使用 10.0.0.0 地址范围。该范围允许采用 10.TE.AM.xx 的 IP 地址格式; 如果使用 B 类或 C 类网络, 只有部分队伍能够遵循这一格式。

How are these addresses assigned? / 这些地址如何分配?

We've covered the basics of what IP addresses are, and which IP addresses we will use for the FRC competition, so now we need to discuss how these addresses will get assigned to the devices on our network. We already stated above that we can't have two devices on the same network with the same IP Address, so we need a way to be sure that every device receives an address without overlapping. This can be done Dynamically (automatic), or Statically (manual).

前面介绍了 IP 地址的基本概念, 以及 FRC 比赛使用的 IP 地址。接下来讨论如何将这些地址分配给网络中的设备。前文已经指出, 同一网络中的两台设备不能使用同一个 IP 地址, 因此需要确保每台设备获得的地址互不重叠。分配方式可以是动态的 (自动), 也可以是静态的 (手动)。

Dynamically / 动态分配

Dynamically assigning IP addresses means that we are letting a device on the network manage the IP address assignments. This is done through the Dynamic Host Configuration Protocol (DHCP). DHCP has many components to it, but for the scope of this document, we will think of it as a service that automatically manages the network. Whenever you plug in a new device to the network, the DHCP service sees the new device, then provides it with an available IP address and the other network settings required for the device to communicate. This can mean that there are times we do not know the exact IP address of each device.

动态分配 IP 地址, 是让网络中的一台设备管理地址分配。这通过动态主机配置协议 (DHCP) 实现。DHCP 包含许多组件, 但在本文范围内, 可以把它理解为自动管理网络的服务。每当新设备接入网络, DHCP 服务就会发现它, 并提供一个可用的 IP 地址及设备通信所需的其他网络设置。这意味着我们有时并不知道每台设备的确切 IP 地址。

What is a DHCP server? / 什么是 DHCP 服务器?

A DHCP server is a device that runs the DHCP service to monitor the network for new devices to configure. In larger businesses, this could be a dedicated computer running the DHCP service and that computer would be the DHCP server. For home networks, FRC networks, and other smaller networks, the DHCP service is usually running on the router; in this case, the router is the DHCP server.

DHCP 服务器是运行 DHCP 服务、监测网络并配置新设备的设备。在大型企业中, 它可能是一台专门运行 DHCP 服务的计算机, 这台计算机就是 DHCP 服务器。在家庭、FRC 及其他较小的网络中, DHCP 服务通常运行在路由器上, 此时路由器就是 DHCP 服务器。

This means that if you ever run into a situation where you need to have a DHCP server assigning IP addresses to your network devices, it's as simple as finding the closest home router, and plugging it in.

因此, 如果遇到需要 DHCP 服务器为网络设备分配 IP 地址的情况, 找一台家用路由器并接入网络即可。

Statically / 静态分配

Statically assigning IP addresses means that we are manually telling each device on the network which IP address we want it to have. This configuration happens through a setting on each device. By disabling DHCP on the network and assigning the addresses manually, we get the benefit of knowing the exact IP address of each device on the network, but because we set each one manually and there is no service keeping track of the used IP addresses, we have to keep track

of this ourselves. While statically setting IP addresses, we must be careful not to assign duplicate addresses, and must be sure we are setting the other network settings (such as subnet mask and default gateway) correctly on each device.

静态分配 IP 地址，是手动指定网络中每台设备应使用的 IP 地址。此配置在各设备的设置中完成。关闭网络中的 DHCP 并手动分配地址，可以明确知道每台设备的确切 IP 地址；但由于地址由我们逐一设置，且没有服务跟踪已用地址，我们必须自行记录。设置静态 IP 地址时，必须注意不要分配重复地址，并确保在每台设备上正确设置其他网络参数，例如子网掩码和默认网关。

What is link-local? / 什么是链路本地地址？

If a device does not have an IP address, then it cannot communicate on a network. This can become an issue if we have a device that is set to dynamically acquire its address from a DHCP server, but there is no DHCP server on the network. An example of this would be when you have a laptop directly connected to a roboRIO and both are set to dynamically acquire an IP address. Neither device is a DHCP server, and since they are the only two devices on the network, they will not be assigned IP addresses automatically.

设备没有 IP 地址就无法在网络上通信。如果设备设为从 DHCP 服务器动态获取地址，而网络中却没有 DHCP 服务器，就会出现问题。例如，笔记本电脑直接连接到 roboRIO，且两者都设置为动态获取 IP 地址。两台设备都不是 DHCP 服务器，而网络中又只有这两台设备，因此它们不会自动获得由 DHCP 分配的 IP 地址。

Link-local addresses give us a standard set of addresses that we can "fall-back" to if a device set to acquire dynamically is not able to acquire an address. If this happens, the device will assign itself an IP address in the **169.254.xx.yy** address range; this is a link-local address. In our roboRIO and computer example above, both devices will realize they haven't been assigned an IP address and assign themselves a link-local address. Once they are both assigned addresses in the **169.254.xx.yy** range, they will be in the same network and will be able to communicate, even though they were set to dynamic and a DHCP server did not assign addresses.

链路本地地址提供一组标准地址，供设置为动态获取地址、但无法获得地址的设备“回退”使用。发生这种情况时，设备会自行分配 **169.254.xx.yy** 范围内的 IP 地址，这就是链路本地地址。在前述 roboRIO 和计算机的例子中，两台设备发现自己未获得 IP 地址后，都会自行分配链路本地地址。当两者都获得 **169.254.xx.yy** 范围内的地址时，它们就处于同一网络，可以互相通信，尽管它们原本设置为动态获取地址，且 DHCP 服务器并未分配地址。

IP Addressing for FRC / FRC 的 IP 地址分配

See the [IP Networking Article](#) for more information.

有关详细信息，请参阅 [IP 网络配置文章](#)。

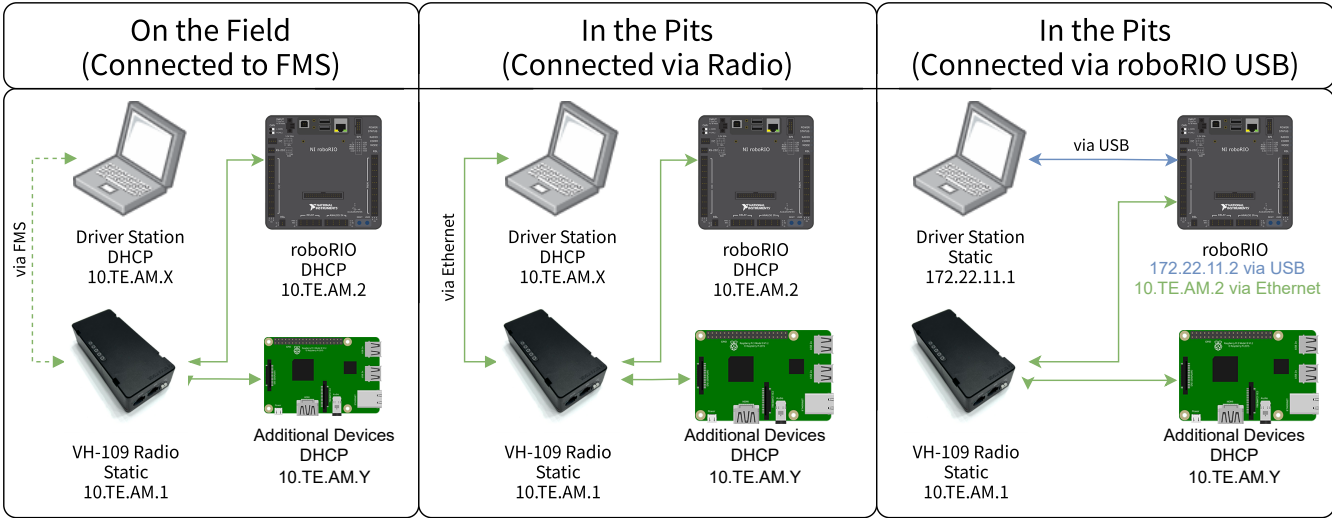
Mixing Dynamic and Static Configurations / 混用动态与静态配置

While on the field, the team should not notice any issues with having devices set statically in the **10.TE.AM.xx** range, and having the field assign DHCP addresses as long as there are no IP address conflicts as referred to in the section above.

在场地上，只要不存在前文所述的 IP 地址冲突，队伍通常不会发现 **10.TE.AM.xx** 范围内的静态设备与场地分配的 DHCP 地址混用所带来的问题。

In the pits, a team may encounter issues with mixing Static and DHCP devices for the following reason. As mentioned above, DHCP devices will fall back to a link-local address (**169.254.xx.yy**) if a server isn't present. For static devices, the IP address will always be the same. If the DHCP server is not present and the roboRIO, driver station, and laptop fall back to link-local addresses, the statically set devices in the **10.TE.AM.xx** range will be in a different network and not visible to those with link-local addresses. A visual description of this is provided below:

在维修区，混用静态设备和 DHCP 设备可能因以下原因导致问题。如前所述，没有服务器时，DHCP 设备会回退到链路本地地址（169.254.xx.yy）。静态设备的 IP 地址则始终相同。如果 DHCP 服务器不存在，而 roboRIO、Driver Station 和笔记本电脑都回退到链路本地地址，那么静态设置在 10.TE.AM.xx 范围内的设备将处于另一网络，使用链路本地地址的设备无法发现它们。下图对此作了直观说明：



How you can't mix link-local and static IP configurations. / 说明链路本地地址与静态 IP 配置不能混用的示意图。

WARNING: When connected via USB to the roboRIO, a Port Forwarding configuration is required to access devices connected to the VH-109 radio (on the green network shown above).

警告： 通过 USB 连接 roboRIO 时，要访问连接到 VH-109 无线电设备的设备（上图绿色网络），必须配置端口转发。

Available Network Ports / 可用的网络端口

Please see R704 of the 2024 Game Manual for information regarding available network ports.

有关可用网络端口的信息，请参阅 2024 年比赛手册的 R704。

译注：原文此处引用 2024 年比赛手册；实际可用端口应以当前赛季手册为准。

mDNS / mDNS

mDNS, or multicast Domain Name System is a protocol that allows us to benefit from the features of DNS, without having a DNS server on the network. To make this clearer, let's take a step back and talk about what DNS is.

mDNS，即多播域名系统，是一种无需在网络中设置 DNS 服务器，也能利用 DNS 功能的协议。为便于理解，先回顾一下什么是 DNS。

What is DNS? / 什么是 DNS?

DNS (Domain Name System) can become a complex topic, but for the scope of this paper, we are going to just look at the high-level overview of DNS. In the most basic explanation, DNS is what allows us to relate human-friendly names for network devices to IP Addresses, and keep track of those IP addresses if they change.

DNS（域名系统）可能是一个复杂的主题，但本文只概述其基本作用。最简单地，DNS 可以将便于人们记忆的网络设备名称关联到 IP 地址，并在地址变化时跟踪这些地址。

Example 1: Let's look at the site www.google.com. The IP address for this site is 172.217.164.132, however that is not very user-friendly to remember!

示例 1：以网站 www.google.com 为例。其 IP 地址是 172.217.164.132，但这个地址并不好记！

Whenever a user types www.google.com into their computer, the computer contacts the DNS server (a setting provided by DHCP!) and asks what is the IP address on file for www.google.com. The DNS server returns the IP address and then the computer is able to use that to connect to the Google website.

用户在计算机中输入 www.google.com 时，计算机将联系 DNS 服务器（其地址由 DHCP 设置提供），查询记录中 www.google.com 对应的 IP 地址。DNS 服务器返回该地址后，计算机就可以用它连接 Google 网站。

Example 2: On your home network, you have a server named [MYCOMPUTER](#) that you want to connect to from your laptop. Your network uses DHCP so you don't know the IP Address of [MYCOMPUTER](#), but DNS allows you to connect just by using the [MYCOMPUTER](#) name. Additionally, whenever the DHCP assignments refresh, [MYCOMPUTER](#) may end up with a different address, but because you're connecting by using the [MYCOMPUTER](#) name instead of a specific IP address, the DNS record was updated and you're still able to connect.

示例 2：家庭网络中有一台名为 [MYCOMPUTER](#) 的服务器，你想从笔记本电脑连接它。网络使用 DHCP，因此你不知道 [MYCOMPUTER](#) 的 IP 地址，但 DNS 允许你只使用名称 [MYCOMPUTER](#) 连接。此外，DHCP 租约更新后，[MYCOMPUTER](#) 可能获得不同地址；但你是通过名称而非特定 IP 地址连接的，DNS 记录也已更新，所以仍然可以连接。

This is the second benefit to DNS and the most relevant for FRC. With DNS, if we reference devices by their friendly name instead of IP Address, we don't have to change anything in our program if the IP Address changes. DNS will keep track of the changes and return the new address if it ever changes.

这是 DNS 的第二个好处，也是与 FRC 最相关的好处。如果用设备的易读名称而不是 IP 地址引用设备，IP 地址变化时就不必修改程序。DNS 会跟踪变化，并在地址发生变化后返回新地址。

DNS for FRC / FRC 中的 DNS

On the field and in the pits, there is no DNS server that allows us to perform the lookups like we do for the Google website, but we'd still like to have the benefits of not remembering every IP Address, and not having to guess at every device's address if DHCP assigns a different address than we expect. This is where mDNS comes into the picture.

场地和维修区都没有像查询 Google 网站那样使用的 DNS 服务器；但我们仍希望不必记住每个 IP 地址，也不必在 DHCP 分配了与预期不同的地址时猜测设备地址。此时就需要 mDNS。

mDNS provides us the same benefits as traditional DNS, but is just implemented in a way that does not require a server. Whenever a user asks to connect to a device using a friendly name, mDNS sends out a message asking the device with that name to identify itself. The device with the name then sends a return message including its IP address so all devices on the network can update their information. mDNS is what allows us to refer to our roboRIO as [roboRIO-TEAM-FRC.local](#) and have it connect on a DHCP network.

mDNS 提供与传统 DNS 相同的便利，但实现方式不需要服务器。用户使用易读名称请求连接某台设备时，mDNS 会发送消息，请求具有该名称的设备表明身份。该设备随后回复包含自身 IP 地址的消息，使网络中的所有设备都可以更新信息。正是 mDNS 让我们能够以 [roboRIO-TEAM-FRC.local](#) 称呼 roboRIO，并在 DHCP 网络上与之连接。

NOTE: If a device used for FRC does not support mDNS, then it will be assigned an IP Address in the 10.TE.AM.20 - 10.TE.AM.255 range, but we won't know the exact address to connect and we won't be able to use the friendly name like before. In this case, the device would need to have a static IP Address.

说明：如果用于 FRC 的设备不支持 mDNS，它会被分配 [10.TE.AM.20](#) - [10.TE.AM.255](#) 范围内的 IP 地址，但我们不知道连接时应使用的确切地址，也无法像前面那样使用易读名称。在这种情况下，该设备需要使用静态 IP 地址。

mDNS - Principles / mDNS——原理

Multicast Domain Name System (mDNS) is a system which allows for resolution of hostnames to IP addresses on small networks with no dedicated name server. To resolve a hostname a device sends out a multicast message to the network querying for the device. The device then responds with a multicast message containing its IP. Devices on the network can store this information in a cache so subsequent requests for this address can be resolved from the cache without repeating the network query.

多播域名系统（mDNS）允许在没有专用名称服务器的小型网络中，将主机名解析为 IP 地址。为解析主机名，设备向网络发送多播消息，查询对应设备。该设备随后通过多播消息回复其 IP 地址。网络中的设备可以缓存这些信息，使后续对该地址的请求直接从缓存解析，而无需重复网络查询。

mDNS - Providers / mDNS——提供程序

To use mDNS, an mDNS implementation is required to be installed on your PC. Here are some common mDNS implementations for each major platform:

使用 mDNS 需要在计算机上安装 mDNS 实现。以下列出各主要平台常见的 mDNS 实现：

Windows:

Windows:

- **NI mDNS Responder:** Installed with the NI FRC Game Tools
- ****NI mDNS Responder:** **随 NI FRC Game Tools 安装。
- **Apple Bonjour:** Installed with iTunes
- ****Apple Bonjour:** **随 iTunes 安装。

OSX:

OSX:

- **Apple Bonjour:** Installed by default
- ****Apple Bonjour:** **默认安装。

Linux:

Linux:

- **nss-mDNS/Avahi/Zeroconf:** Installed and enabled by default on some Linux variants (such as Ubuntu or Mint). May need to be installed or enabled on others (such as Arch)
- ****nss-mDNS/Avahi/Zeroconf:** **某些 Linux 发行版（如 Ubuntu 或 Mint）默认安装并启用；其他发行版（如 Arch）可能需要自行安装或启用。

mDNS - Firewalls / mDNS——防火墙

NOTE: Depending on your PC configuration, no changes may be required, this section is provided to assist with troubleshooting.

说明： 根据计算机配置，可能无需修改任何设置。本节用于协助排障。

To work properly mDNS must be allowed to pass through your firewall. Because the network traffic comes from the mDNS implementation and not directly from the Driver Station or IDE, allowing those applications through may not be sufficient.

There are two main ways to resolve mDNS firewall issues:

mDNS 要正常工作，网络流量必须能够通过防火墙。由于流量来自 mDNS 实现，而不是直接来自 Driver Station 或 IDE，仅放行这些应用可能不够。解决 mDNS 防火墙问题主要有两种方式：

- Add an application/service exception for the mDNS implementation (NI mDNS Responder is `C:\Program Files\National Instruments\Shared\mDNS Responder\nimdnsResponder.exe`)
- 为 mDNS 实现添加应用 / 服务例外（NI mDNS Responder 位于 `C:\Program Files\National Instruments\Shared\mDNS Responder\nimdnsResponder.exe`）。
- Add a port exception for traffic to/from UDP 5353. IP Ranges:
- 为进出 UDP 5353 的流量添加端口例外。IP 范围：
 - `10.0.0.0 - 10.255.255.255`
 - `172.16.0.0 - 172.31.255.255`
 - `192.168.0.0 - 192.168.255.255`
 - `169.254.0.0 - 169.254.255.255`
 - `224.0.0.251`

mDNS - Browser support / mDNS——浏览器支持

Most web-browsers should be able to utilize the mDNS address to access the roboRIO web server as long as an mDNS provider is installed. These browsers include Microsoft Edge, Firefox, and Google Chrome.

只要安装了 mDNS 提供程序，大多数 Web 浏览器就应当能够使用 mDNS 地址访问 roboRIO Web 服务器。这些浏览器包括 Microsoft Edge、Firefox 和 Google Chrome。

USB / USB

If using the USB interface, no network setup is required, but you do need the FRC Game Tools installed to provide the roboRIO USB Driver. The roboRIO driver will automatically configure the IP address of the host (your computer) and roboRIO and the software listed above should be able to locate and utilize your roboRIO.

使用 USB 接口时无需设置网络，但必须安装 FRC Game Tools，以提供 roboRIO USB 驱动程序。roboRIO 驱动程序会自动配置主机（你的计算机）和 roboRIO 的 IP 地址，上述软件随后应当能够找到并使用 roboRIO。

Ethernet/Wireless / 以太网 / 无线连接

The Programming your Radio will enable the DHCP server on the VH-109 radios. Both Robot Radio (`10.TE.AM.1`) and AP Radio (`10.TE.AM.4`) will have the team-based IP address and will hand out DHCP address from `10.TE.AM.20` to `10.TE.AM.199`. When connected to the field, FMS will also hand out addresses in the same IP range.

无线电设备编程会启用 VH-109 无线电设备上的 DHCP 服务器。机器人无线电设备 (`10.TE.AM.1`) 和 AP 无线电设备 (`10.TE.AM.4`) 都会使用基于队伍号码的 IP 地址，并分配 `10.TE.AM.20` 至 `10.TE.AM.199` 的 DHCP 地址。连接到场地时，FMS 也会在同一 IP 范围内分配地址。

Summary / 总结

IP Addresses are what allow us to communicate with devices on a network. For FRC, these addresses are going to be in the `10.TE.AM.xx` range if we are connected to a DHCP server or if they are assigned statically, or in the link-local

169.254.xx.yy range if the devices are set to DHCP, but there is no server present. For more information on how IP Addresses work, see [this](#) article by Microsoft.

IP 地址使我们能够与网络设备通信。对 FRC 而言，若连接到 DHCP 服务器或使用静态分配，地址将处于 10.TE.AM.xx 范围；若设备设置为 DHCP，但网络中没有服务器，则地址会处于链路本地的 169.254.xx.yy 范围。有关 IP 地址工作方式的更多信息，请参阅 Microsoft 的[这篇文章](#)。

If all devices on the network support mDNS, then all devices can be set to DHCP and referred to using their friendly names (ex. roboRIO-TEAM-FRC.local). If some devices do not support mDNS, they will need to be set to use static addresses.

如果网络中的所有设备都支持 mDNS，就可以将它们全部设为 DHCP，并通过易读名称引用（例如 roboRIO-TEAM-FRC.local）。不支持 mDNS 的设备则需要设置静态地址。

If all devices are set to use DHCP or Static IP assignments (with correct static settings), the communication should work in both the pit and on the field without any changes needed. If there are a mix of some Static and some DHCP devices, then the Static devices will connect on the field, but will not connect in the pit. This can be resolved by either setting all devices to static settings, or leaving the current settings and providing a DHCP server in the pit.

如果所有设备都使用 DHCP 或静态 IP 分配（且静态设置正确），在维修区和场地上通信应当无需修改配置即可正常工作。如果部分设备为静态配置，部分为 DHCP 配置，那么静态设备在场地上可以连接，但在维修区无法连接。可以将所有设备都改为静态配置，或者保持当前设置并在维修区提供 DHCP 服务器，来解决此问题。

IP Configurations / IP 配置

This document describes the IP configuration used at events, both on the fields and in the pits, potential issues, and workaround configurations.

本文介绍赛事场地和维修区使用的 IP 配置、潜在问题及变通配置。

TE.AM IP Address Notation / TE.AM IP 地址写法

The notation TE.AM is used as part of IPs in numerous places. This notation refers to splitting your five digit team number into digits for the IP address octets. Where AM is the last two digits of the team number, and TE is the first three digits. Leading zeros are optional. This scheme supports team numbers up to 25599.

许多地方使用 TE.AM 作为 IP 地址的一部分。这种写法将五位队伍号码拆分，用于 IP 地址的八位字节：AM 是队伍号码的最后两位，TE 是前三位。前导零可以省略。此方案支持最高 25599 的队伍号码。

Example: 10.TE.AM.2

示例: 10.TE.AM.2

Team 1 - 10.0.1.2

队伍 1 - 10.0.1.2

Team 12 - 10.0.12.2

队伍 12 - 10.0.12.2

Team 122 - 10.1.22.2

队伍 122 - 10.1.22.2

Team 1002 - 10.10.2.2

队伍 1002 - 10.10.2.2

Team 1212 - 10.12.12.2

队伍 1212 - 10.12.12.2

Team 1202 - 10.12.2.2

队伍 1202 - 10.12.2.2

Team 1220 - 10.12.20.2

队伍 1220 - 10.12.20.2

Team 3456 - 10.34.56.2

队伍 3456 - 10.34.56.2

Team 10000 - 10.100.0.2

队伍 10000 - 10.100.0.2

Team 12345 - 10.123.45.2

DHCP configuration / DHCP 配置

This section describes networking when using DHCP to assign IP addresses on the field, in the pits, and at home.

本节介绍在场地、维修区和家中使用 DHCP 分配 IP 地址时的网络配置。

On the Field DHCP Configuration / 场地上的 DHCP 配置

Both the Field Network and Robot Radio run DHCP servers with non-overlapping pools. The Field Network DHCP server provides an IP address for the Driver Station, and the Robot Radio provides IP addresses for devices onboard the robot. The IP ranges, subnet mask, and default gateway are listed in the table below.

场地网络和机器人无线电设备都运行 DHCP 服务器，且两者的地址池互不重叠。场地网络的 DHCP 服务器为 Driver Station 提供 IP 地址，机器人无线电设备则为机器人上的设备提供 IP 地址。下表列出 IP 地址范围、子网掩码和默认网关。

	Field Network / 场地网络	Robot Radio / 机器人无线电设备
DHCP Range / DHCP 范围	10.TE.AM.20 to 10.TE.AM.199 / 10.TE.AM.20 至 10.TE.AM.199	10.TE.AM.200 to 10.TE.AM.219 / 10.TE.AM.200 至 10.TE.AM.219
Subnet Mask / 子网掩码	255.255.255.0	255.255.255.0
Default Gateway / 默认网关	10.TE.AM.4	10.TE.AM.4 (10.TE.AM.1 if using the OpenMesh radio) / 10.TE.AM.4 (使用 OpenMesh 无线电设备时为 10.TE.AM.1)

- Vivid-Hosting VH-109 Robot Radio - Static 10.TE.AM.1 programmed by Kiosk
- Vivid-Hosting VH-109 机器人无线电设备 - 由 Kiosk 编程为静态地址 10.TE.AM.1。
- roboRIO - DHCP 10.TE.AM.2 assigned by the Robot Radio
- roboRIO - 由机器人无线电设备通过 DHCP 分配 10.TE.AM.2。
- Driver Station - DHCP ("Obtain an IP address automatically") 10.TE.AM.X assigned by field
- Driver Station - 由场地通过 DHCP (“自动获取 IP 地址”) 分配 10.TE.AM.X。
- Other devices (if used) - DHCP 10.TE.AM.YY assigned by Robot Radio
- 其他设备（如使用）- 由机器人无线电设备通过 DHCP 分配 10.TE.AM.YY。

In the Pits DHCP Configuration / 维修区中的 DHCP 配置

- VH-109 radio - Static 10.TE.AM.1 programmed by Kiosk.
- VH-109 无线电设备 - 由 Kiosk 编程为静态地址 10.TE.AM.1。
- roboRIO - 10.TE.AM.2, assigned by Robot Radio
- roboRIO - 由机器人无线电设备分配 10.TE.AM.2。
- Driver Station - DHCP ("Obtain an IP address automatically"), 10.TE.AM.X, assigned by Robot Radio
- Driver Station - 由机器人无线电设备通过 DHCP (“自动获取 IP 地址”) 分配 10.TE.AM.X。
- Other devices (if used) - DHCP, 10.TE.AM.Y, assigned by Robot Radio
- 其他设备（如使用）- 由机器人无线电设备通过 DHCP 分配 10.TE.AM.Y。

NOTE: If the roboRIO is not being assigned a DHCP IP address of **10.TE.AM.2**, ensure you've Set the roboRIO team number

说明: 如果 roboRIO 未通过 DHCP 获得 **10.TE.AM.2**, 请确认已设置 roboRIO 队伍号码。

At Home DHCP Configuration / 家中的 DHCP 配置

- VH-109 radio - Static **10.TE.AM.1** programmed when configuring robot radio
- VH-109 无线电设备 - 配置机器人无线电设备时编程为静态地址 **10.TE.AM.1**。
- roboRIO - **10.TE.AM.2**, assigned by Robot Radio
- roboRIO - 由机器人无线电设备分配 **10.TE.AM.2**。
- VH-109 access point radio - Static **10.TE.AM.4**, programmed when configuring access point radio
- VH-109 接入点无线电设备 - 配置接入点无线电设备时编程为静态地址 **10.TE.AM.4**。
- Driver Station - DHCP ("Obtain an IP address automatically"), **10.TE.AM.X**, assigned by Robot Radio
- Driver Station - 由机器人无线电设备通过 DHCP ("自动获取 IP 地址") 分配 **10.TE.AM.X**。
- Other devices (if used) - DHCP, **10.TE.AM.Y**, assigned by Robot Radio
- 其他设备 (如使用) - 由机器人无线电设备通过 DHCP 分配 **10.TE.AM.Y**。

NOTE: If the roboRIO is not being assigned a DHCP IP address of **10.TE.AM.2**, ensure you've Set the roboRIO team number

说明: 如果 roboRIO 未通过 DHCP 获得 **10.TE.AM.2**, 请确认已设置 roboRIO 队伍号码。

Static Configuration / 静态配置

It is also possible to configure static IPs on your devices to accommodate devices or software which do not support mDNS. When doing so you want to make sure to avoid addresses that will be in use when the robot is on the field network. These addresses are **10.TE.AM.1** for the VH-109 radio, **10.TE.AM.3** and **10.TE.AM.4** for the field network, and anything **10.TE.AM.20** or greater which may be assigned to a device configured for DHCP or else reserved. The roboRIO network configuration can be set from the webdashboard.

为适配不支持 mDNS 的设备或软件, 也可以在设备上配置静态 IP。这样做时, 应避免机器人连接场地网络后会被占用的地址: VH-109 无线电设备使用 **10.TE.AM.1**; 场地网络使用 **10.TE.AM.3** 和 **10.TE.AM.4**; **10.TE.AM.20** 及以上的地址可能分配给 DHCP 设备, 也可能另有保留。roboRIO 的网络配置可以通过 webdashboard 设置。

- VH-109 radio - Static **10.TE.AM.1** programmed when configuring robot radio
- VH-109 无线电设备 - 配置机器人无线电设备时编程为静态地址 **10.TE.AM.1**。
- roboRIO - Static **10.TE.AM.2** subnet mask of **255.255.255.0** (default)
- roboRIO - 静态地址 **10.TE.AM.2**, 子网掩码为 **255.255.255.0** (默认值)。
- VH-109 access point radio - Static **10.TE.AM.4**, programmed when configuring access point radio
- VH-109 接入点无线电设备 - 配置接入点无线电设备时编程为静态地址 **10.TE.AM.4**。
- Driver Station - Static **10.TE.AM.5** subnet mask **must** be **255.0.0.0** to enable the DS to reach both the robot and FMS Server, without additionally configuring the default gateway. If a static address is assigned and the subnet mask is set to **255.255.255.0**, then the default gateway must be configured to **10.TE.AM.4**.
- Driver Station - 静态地址 **10.TE.AM.5**; 若不额外配置默认网关, 子网掩码**必须**为 **255.0.0.0**, 才能使 DS 同时访问机器人和 FMS 服务器。如果设置了静态地址, 但子网掩码设为 **255.255.255.0**, 则必须将默认网关配置为 **10.TE.AM.4**。

- Other devices - Static 10.10.10.6-.19 subnet 255.255.255.0
- 其他设备 - 静态地址 10.10.10.6-.19, 子网掩码 255.255.255.0。

roboRIO Network Troubleshooting / roboRIO 网络故障排查

The roboRIO and FRC® tools use dynamic IP addresses (DHCP) for network connectivity. This article describes steps for troubleshooting networking connectivity between your PC and your roboRIO

roboRIO 和 FRC® 工具使用动态 IP 地址 (DHCP) 进行网络连接。本文介绍排查计算机与 roboRIO 之间网络连接问题的步骤。

Ping the roboRIO using mDNS / 使用 mDNS ping roboRIO

The first step to identifying roboRIO networking issues is to isolate if it is an application issue or a general network issue. To do this, click **Start -> type cmd -> press Enter** to open the command prompt. Type `ping roboRIO-#####-FRC.local` where ##### is your team number (with no leading zeroes) and press enter. If the ping succeeds, the issue is likely with the specific application, verify your team number configuration in the application, and check your firewall configuration.

排查 roboRIO 网络问题的第一步，是区分应用程序问题和一般网络问题。为此，点击**开始 -> 输入 cmd -> 按 Enter**，打开命令提示符。输入 `ping roboRIO-#####-FRC.local`，其中 ##### 是不带前导零的队伍号码，然后按 Enter。如果 ping 成功，问题可能出在特定应用程序上；请核对应用程序中的队伍号码配置，并检查防火墙配置。

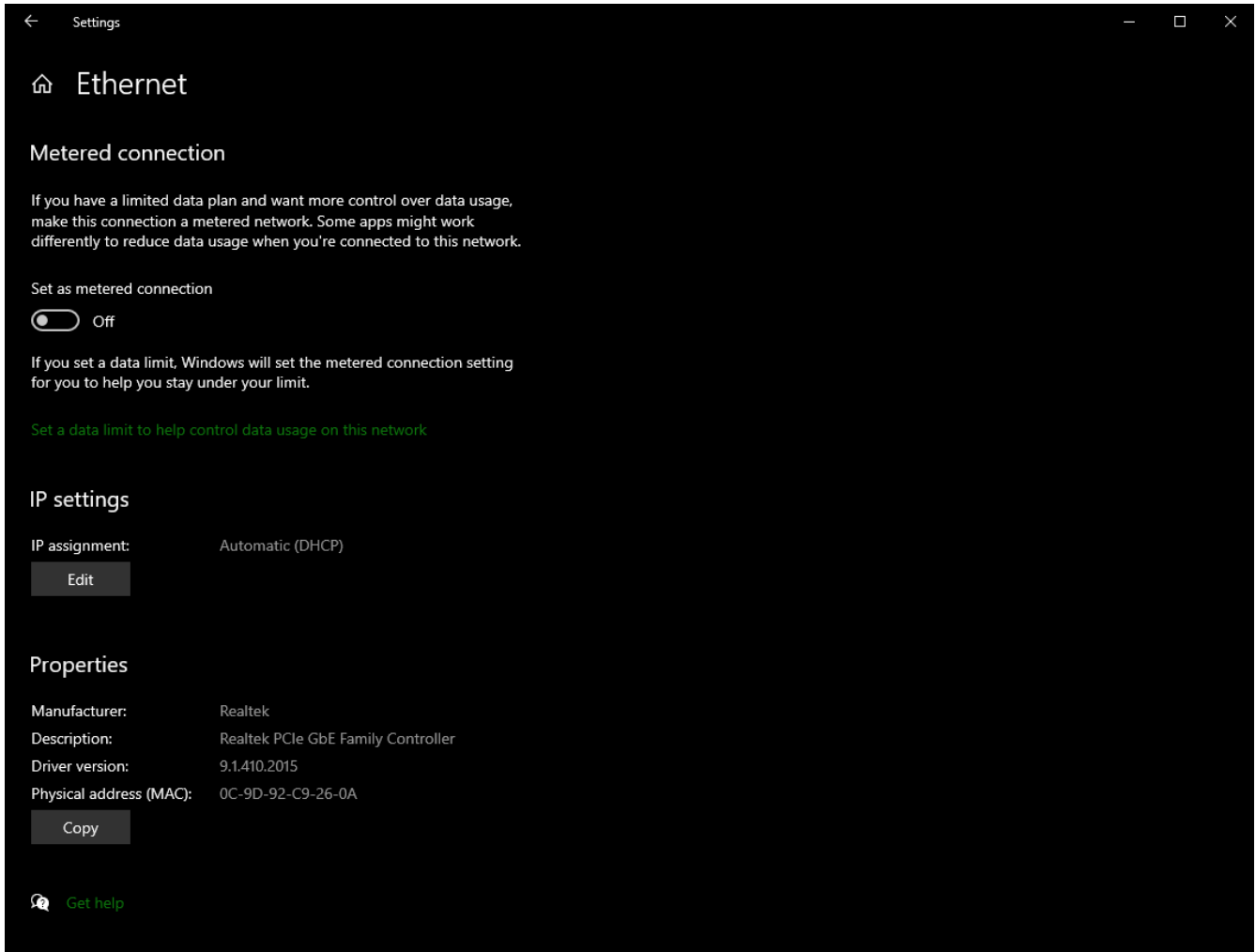
Ping the roboRIO IP Address / ping roboRIO 的 IP 地址

If there is no response, try pinging `10.TE.AM.2` ([TE.AM IP Notation](#)) using the command prompt as described above. If this works, you have an issue resolving the mDNS address on your PC. The two most common causes are not having an mDNS resolver installed on the system and a DNS server on the network that is trying to resolve the .local address using regular DNS.

如果没有响应，按上文所述在命令提示符中尝试 ping `10.TE.AM.2` (参阅 [TE.AM IP 地址写法](#))。如果成功，说明计算机无法解析 mDNS 地址。最常见的两种原因是系统未安装 mDNS 解析器，或网络中的 DNS 服务器正尝试使用普通 DNS 解析 .local 地址。

- Verify that you have an mDNS resolver installed on your system. On Windows, this is typically fulfilled by the NI FRC Game Tools. For more information on mDNS resolvers, see the [Network Basics article](#).
- 确认系统已安装 mDNS 解析器。在 Windows 上，NI FRC Game Tools 通常会提供该功能。有关 mDNS 解析器的更多信息，请参阅[网络基础文章](#)。
- Disconnect your computer from any other networks. Removing any other routers from the system will help verify that there is not a DNS server causing the issue.
- 断开计算机与其他网络的连接。移除系统中的其他路由器，有助于确认问题并非由 DNS 服务器导致。

Ping Fails / ping 失败



Windows 10+ image of the adapter setting / Windows 10+ 的网络适配器设置截图。

If pingging the IP address directly fails, you may have an issue with the network configuration of the PC. The PC should be configured to **Automatic**. To check this, click **Start -> Settings -> Network & Internet**. Depending on your network, select **Wifi** or **Ethernet**. Then click on your connected network. Scroll down to **IP settings** and click **Edit** and ensure the **Automatic (DHCP)** option is selected.

如果直接 ping IP 地址也失败，计算机的网络配置可能有问题。计算机应配置为**自动**获取地址。检查方法：点击**开始 -> 设置 -> 网络和 Internet**。根据所用网络选择 **Wifi** 或**以太网**，然后点击已连接的网络。向下滚动到 **IP 设置**，点击**编辑**，确保选择了**自动 (DHCP)**。

USB Connection Troubleshooting / USB 连接故障排查

If you are attempting to troubleshoot the USB connection, try pingging the roboRIO's IP address. As long as there is only one roboRIO connected to the PC, it should be configured as 172.22.11.2. If this ping fails, make sure you have the roboRIO connected and powered, and that you have installed the NI FRC Game Tools. The game tools installs the roboRIO drivers needed for the USB connection.

排查 USB 连接时，请尝试 ping roboRIO 的 IP 地址。只要计算机仅连接了一台 roboRIO，其地址应配置为 **172.22.11.2**。如果 ping 失败，请确认 roboRIO 已连接并通电，且已安装 NI FRC Game Tools。该工具会安装 USB 连接所需的 roboRIO 驱动程序。

If this ping succeeds, but the .local ping fails, it is likely that either the roboRIO hostname is configured incorrectly, or you are connected to a DNS server which is attempting to resolve the .local address.

如果此 ping 成功，但对 `.local` 地址的 ping 失败，那么 roboRIO 主机名可能配置错误，或者计算机连接了某个正尝试解析 `.local` 地址的 DNS 服务器。

- Verify that your roboRIO has been imaged for your team number: roboRIO 1 roboRIO 2. This sets the hostname used by mDNS.
- 确认 roboRIO 已按本队号码刷写镜像：roboRIO 1、roboRIO 2。此过程会设置 mDNS 使用的主机名。
- Disable all other network adapters
- 禁用所有其他网络适配器。

Ethernet Connection / 以太网连接

The IP address from the roboRIO webdashboard. / roboRIO webdashboard 中显示的 IP 地址。

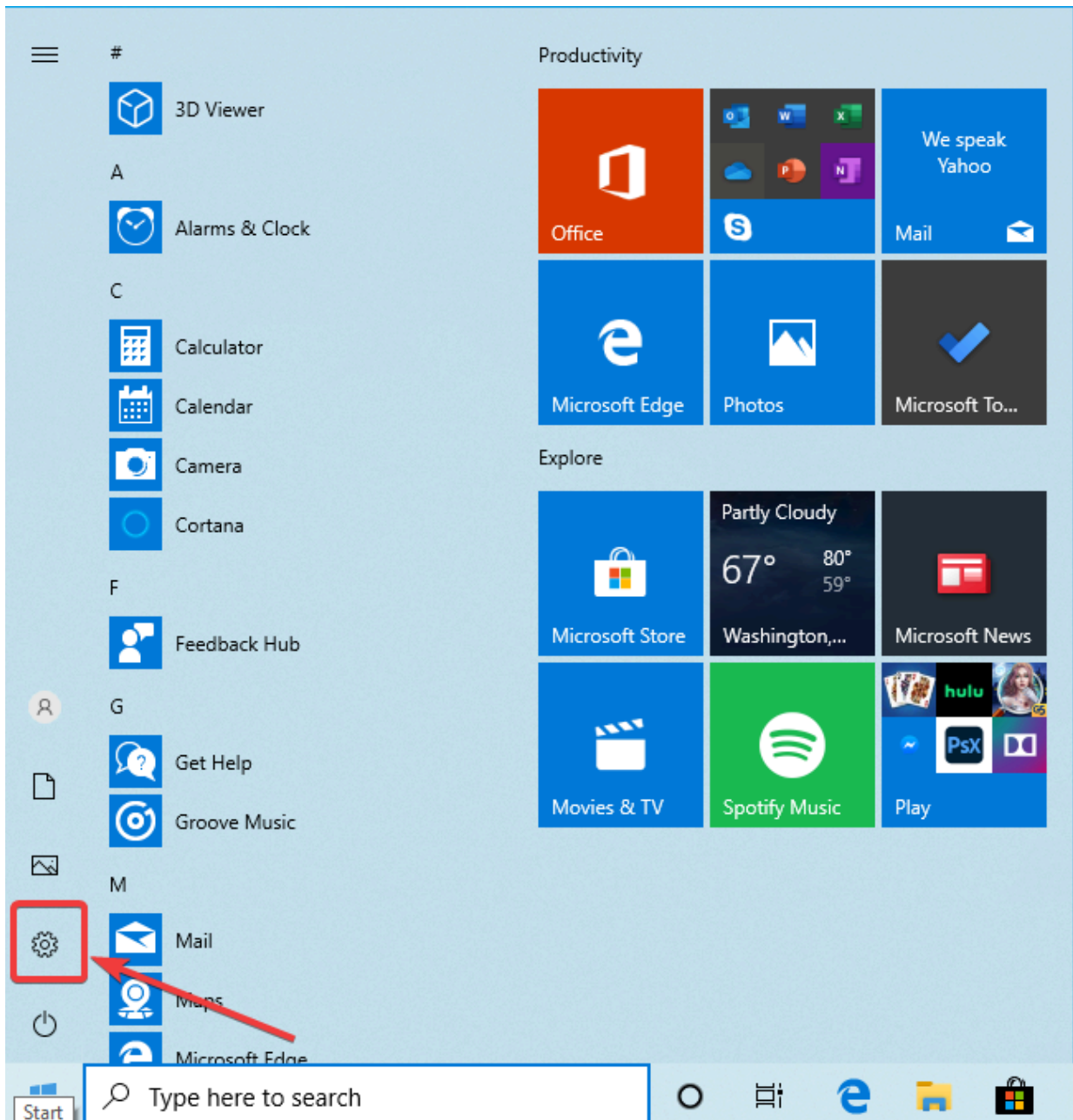
If you are troubleshooting an Ethernet connection, it may be helpful to first make sure that you can connect to the roboRIO using the USB connection. Using the USB connection, open the roboRIO webdashboard and verify that the roboRIO has an IP address on the ethernet interface. If you are tethering to the roboRIO directly this should be a self-assigned `169.*.*.*` address, if you are connected to the VH-109 radio, it should be `10.TE.AM.2` where TEAM is your five digit FRC team number([TE.AM IP Notation](#)). If the only IP address here is the USB address, verify the physical roboRIO ethernet connection. If the IP address is `10.TE.AM.XX`, ensure you've set the roboRIO team number

排查以太网连接时，先确认能否通过 USB 连接 roboRIO，可能会有所帮助。通过 USB 打开 roboRIO webdashboard，确认 roboRIO 的以太网接口有 IP 地址。如果计算机直接连接 roboRIO，应显示自行分配的 `169.*.*.*` 地址；如果连接到 VH-109 无线电设备，应显示 `10.TE.AM.2`，其中 TEAM 是五位 FRC 队伍号码（参阅 [TE.AM IP 地址写法](#)）。如果这里只有 USB 地址，请检查 roboRIO 以太网的物理连接。如果 IP 地址是 `10.TE.AM.XX`，请确认已设置 roboRIO 队伍号码。

Disabling Network Adapters / 禁用网络适配器

This is not always the same as turning the adapters off with a physical button or putting the PC into airplane mode. The following steps provide more detail on how to disable adapters.

这与通过物理按钮关闭适配器，或将计算机设为飞行模式，并不总是一回事。以下步骤详细说明如何禁用适配器。



Windows 10+ settings gear on the left hand side of the start menu. / Windows 10+ 开始菜单左侧的“设置”齿轮图标。

Open the Settings application by clicking on the settings icon.

点击“设置”图标，打开“设置”应用。

Windows Settings

**System**Display, sound, notifications,
power**Devices**

Bluetooth, printers, mouse

**Phone**

Link your Android, iPhone

**Network & Internet**

Wi-Fi, airplane mode, VPN

**Personalization**

Background, lock screen, colors

**Apps**Uninstall, defaults, optional
features**Accounts**Your accounts, email, sync,
work, family**Time & Language**

Speech, region, date

**Gaming**Xbox Game Bar, captures, Game
Mode**Ease of Access**Narrator, magnifier, high
contrast**Search**

Find my files, permissions

**Privacy**

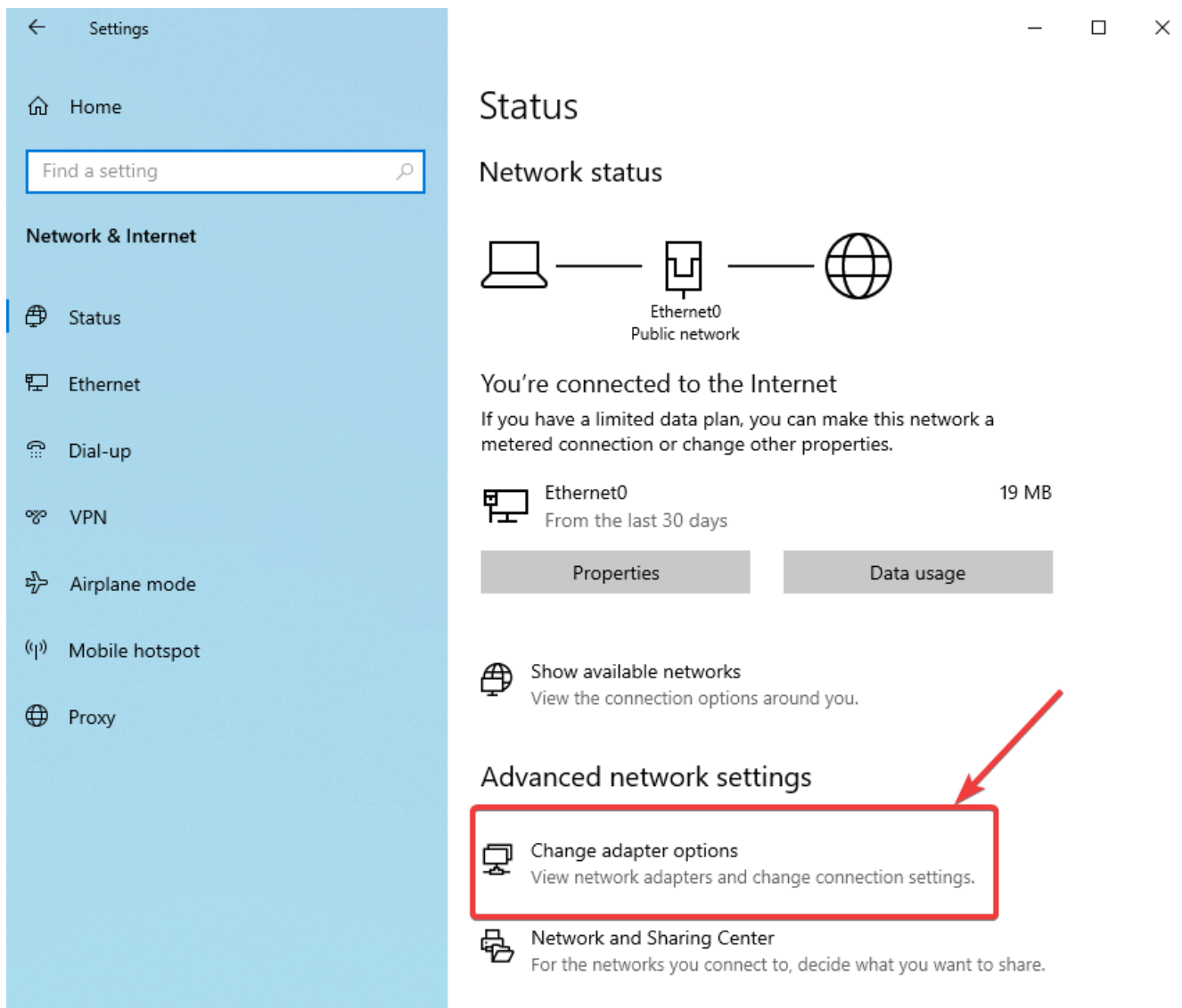
Location, camera, microphone

**Update & Security**Windows Update, recovery,
backup

Clicking the "Network & Internet" category under Windows Settings. / 在 Windows 设置中点击“网络和 Internet”类别。

Choose the **Network & Internet** category.

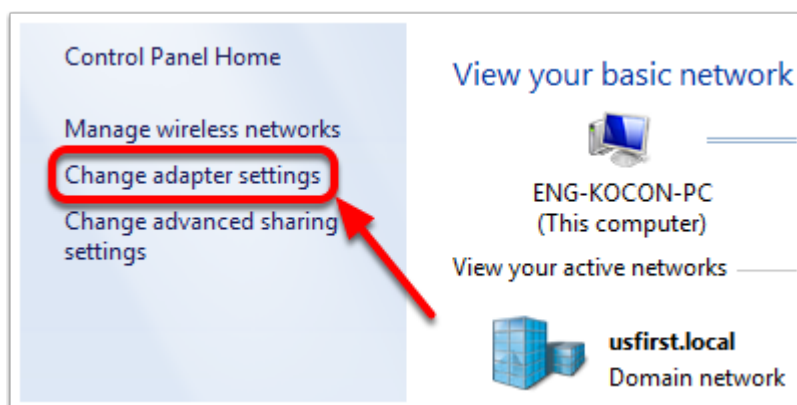
选择**网络和 Internet**类别。



Choose "Change adapter options" in the bottom of the screen. / 选择屏幕底部的“更改适配器选项”。

Click on **Change adapter options**.

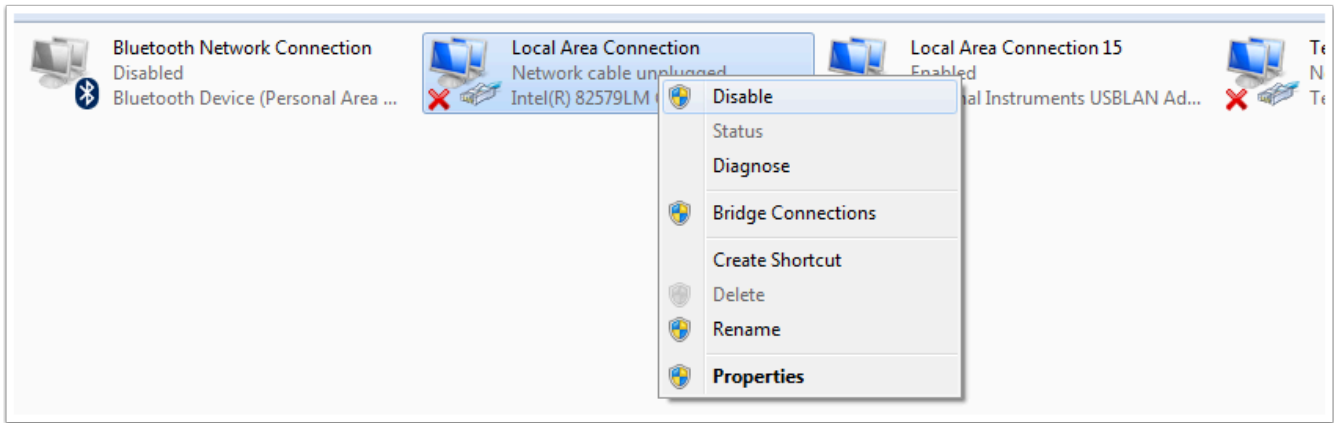
点击**更改适配器选项**。



Click "change adapter settings" in the bar on the left hand side. / 点击左侧栏的“更改适配器设置”。

On the left pane, click **Change Adapter Settings**.

在左侧窗格中，点击**更改适配器设置**。



Right clicking on the adapter in question and choosing the first option "Disable". / 右键点击相关适配器并选择第一个选项“禁用”。

For each adapter other than the one connected to the radio, right click on the adapter and select **Disable** from the menu.

对于连接无线电设备的适配器以外的每个适配器，右键点击该适配器，并从菜单中选择**禁用**。

Proxies / 代理

- Proxies. Having a proxy enabled may cause issues with the roboRIO networking.
- 代理。启用代理可能导致 roboRIO 网络连接问题。

Windows Firewall Configuration / Windows 防火墙配置

Many of the programming tools used in FRC® need network access for various reasons. Depending on the exact configuration, the Windows Firewall may potentially interfere with this access for one or more of these programs.

FRC® 使用的许多编程工具因各种原因需要网络访问。根据具体配置，Windows 防火墙可能干扰其中一个或多个程序的网络访问。

Disabling Windows Firewall / 禁用 Windows 防火墙

IMPORTANT: Disabling your firewall requires administrator privileges to the PC. Additionally note that disabling the firewall is not recommended for computers that connect to the internet.

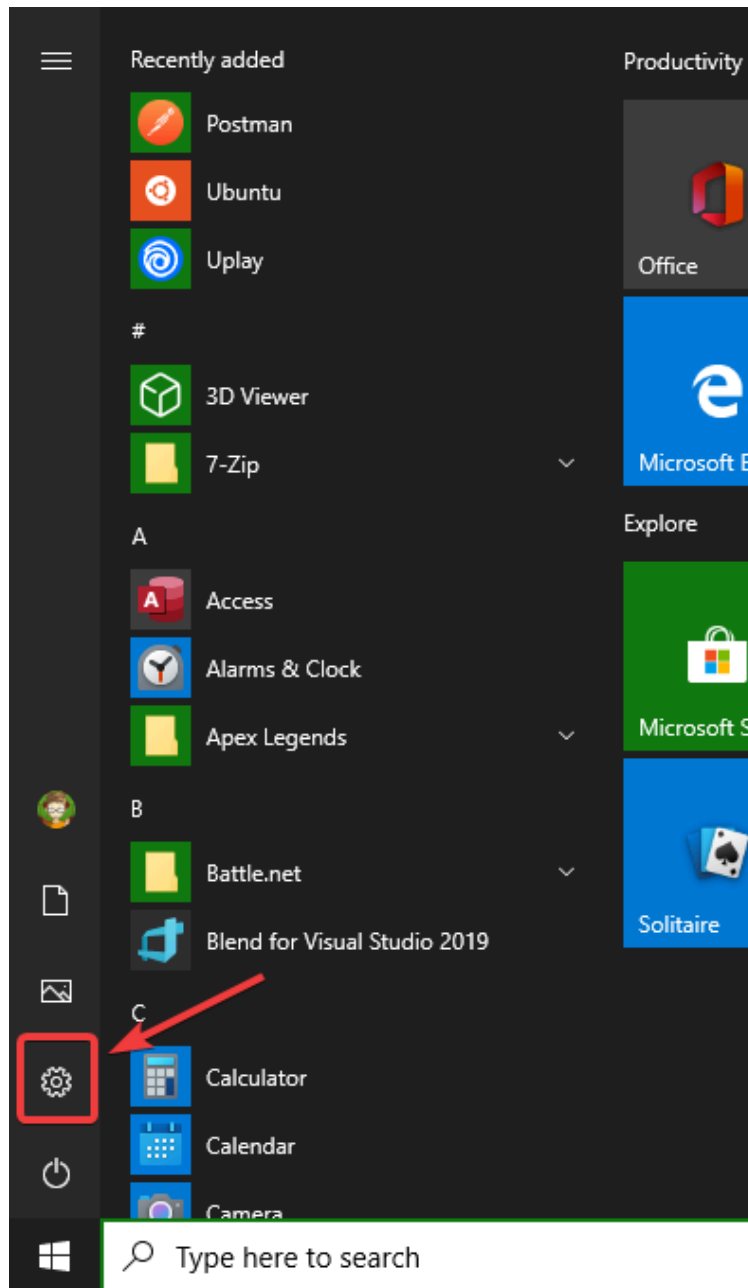
重要： 禁用防火墙需要计算机管理员权限。另请注意，不建议在连接互联网的计算机上禁用防火墙。

The easiest solution is to disable the Windows Firewall. Teams should beware that this does make the PC potentially more vulnerable to malware attacks if connecting to the internet.

最简单的解决方法是禁用 Windows 防火墙。队伍应注意，如果计算机连接互联网，这可能使其更容易受到恶意软件攻击。

Click **Start** -> **Settings**

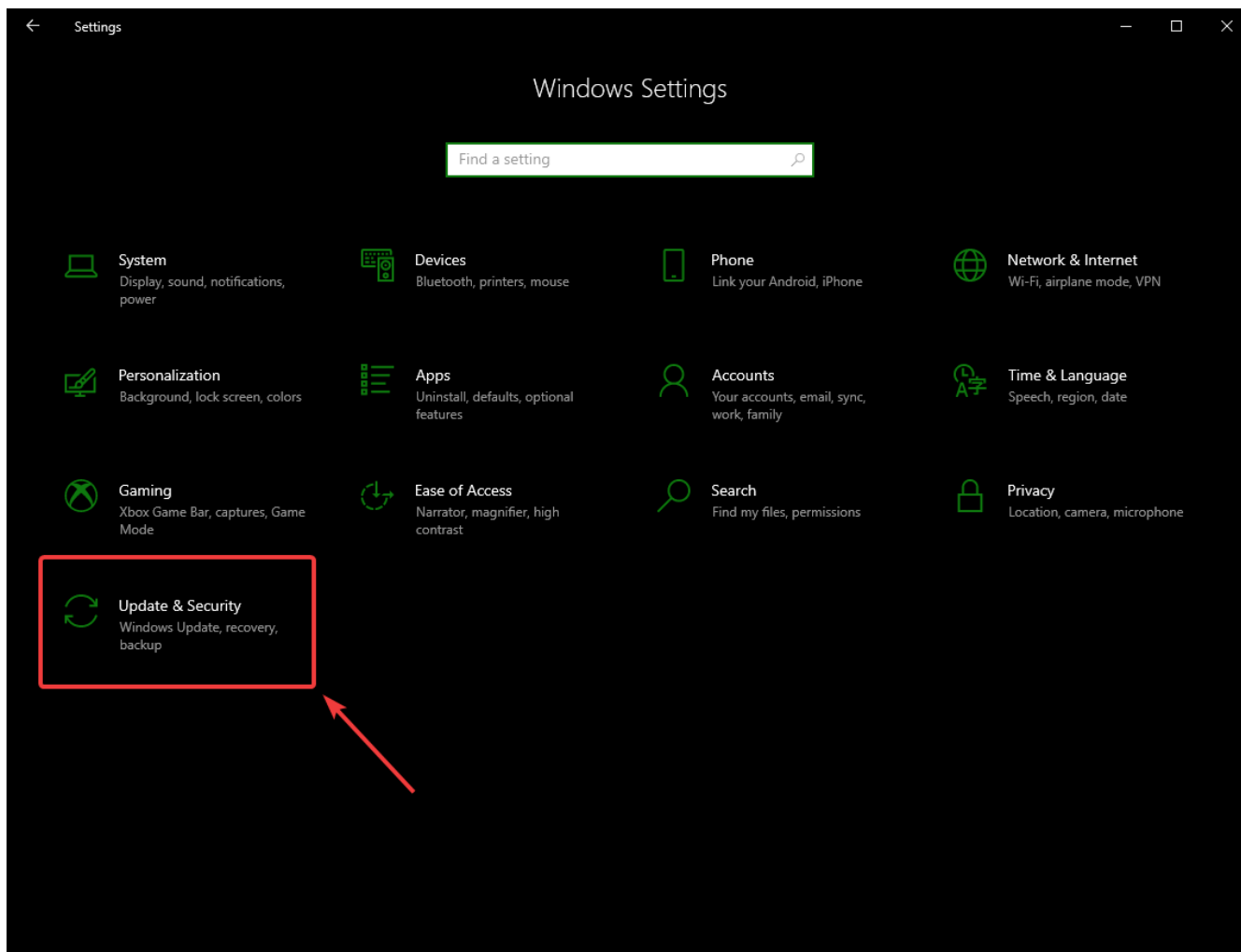
点击**开始** -> **设置**。



Setting gear icon under the start menu. / 开始菜单中的“设置”齿轮图标。

Click **Update & Security**

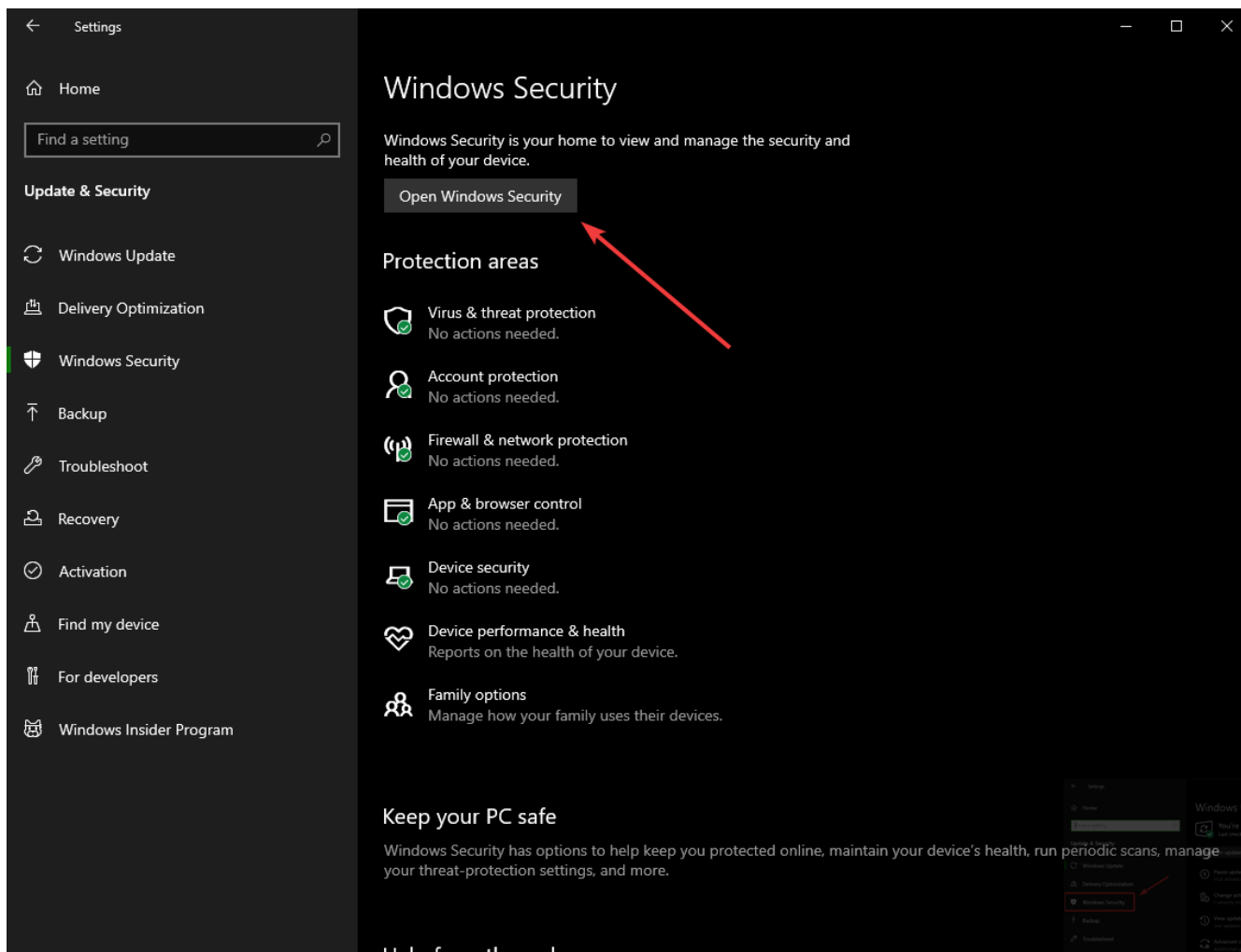
点击**更新和安全**。



Click the "Update & Security" category. / 点击“更新和安全”类别。

In the right pane, select **Open Windows Security**

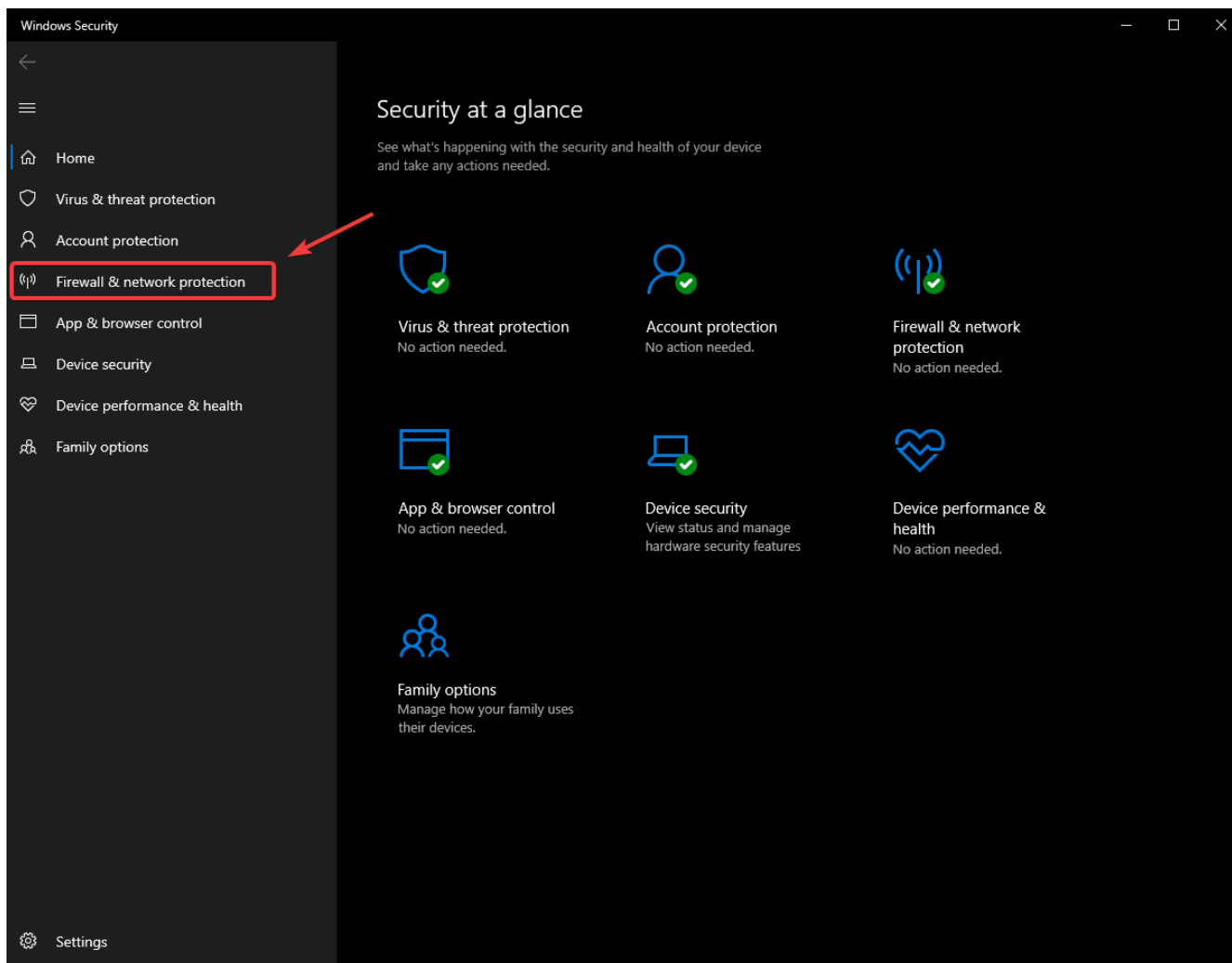
在右侧窗格中，选择**打开 Windows 安全中心**。



Click the "Open Windows Security" button at the top of the right pane. / 点击右侧窗格顶部的“打开 Windows 安全中心”按钮。

In the left pane, select **Firewall and network protection**

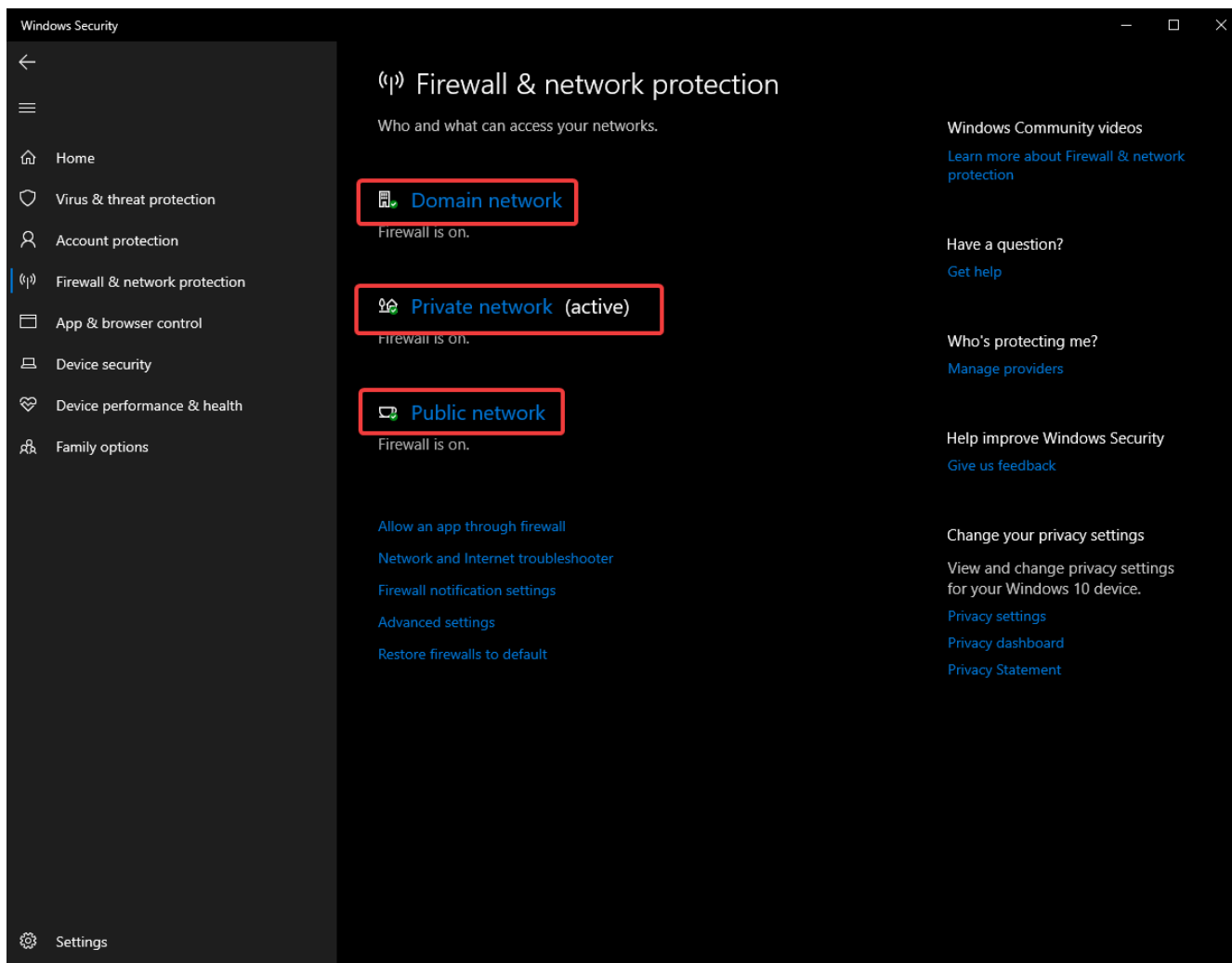
在左侧窗格中，选择**防火墙和网络保护**。



Left pane option on the Security at a glance page. / “安全性概览”页面左侧窗格中的选项。

Click on **each** of the highlighted options

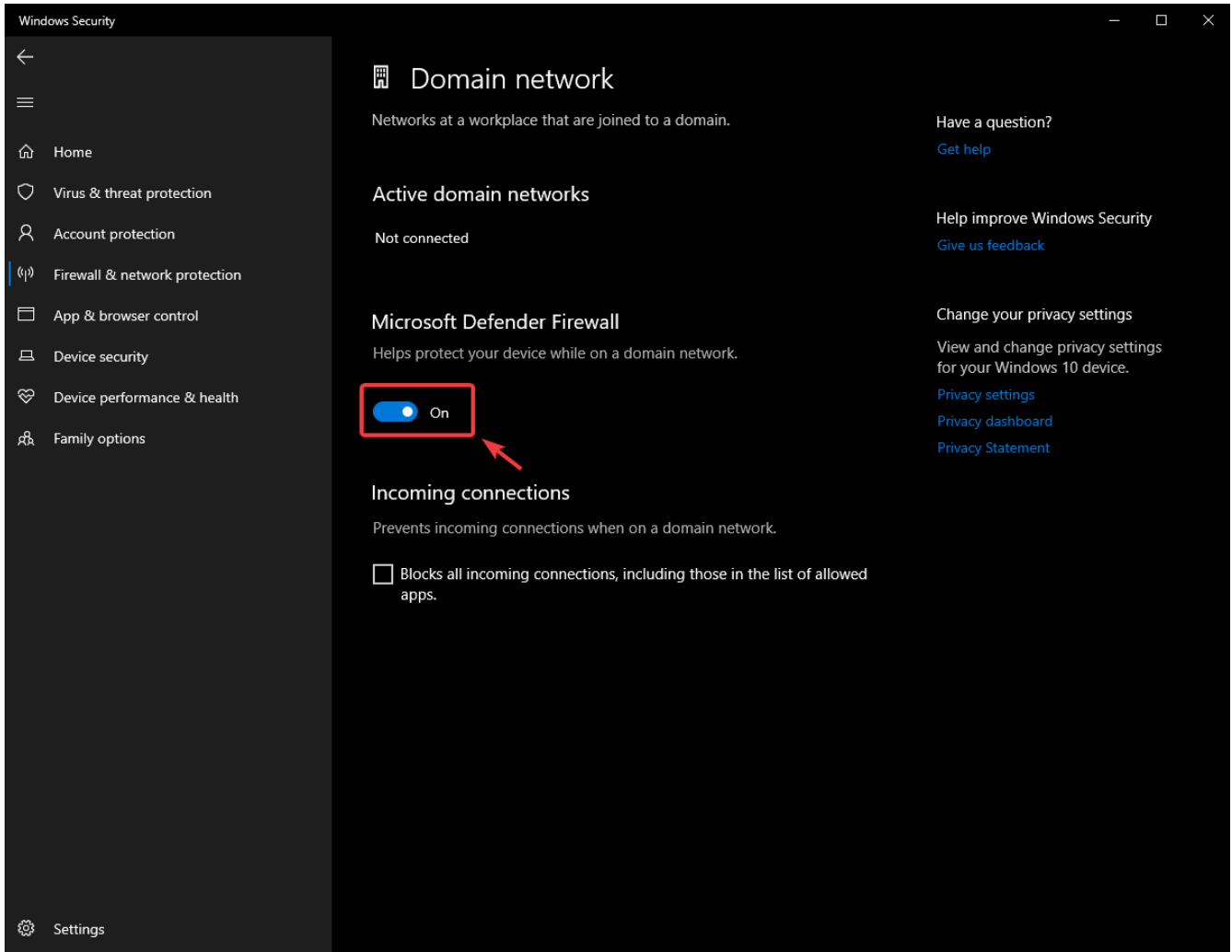
点击突出显示的**每个**选项。



Click on the three highlighted options: Domain network, Private network, and Public Network. / 点击突出显示三个选项：“域网络”“专用网络”和“公用网络”。

Then click on the **On** toggle to turn it off.

然后点击**开**切换按钮，将其关闭。



The toggle button in the center for each option. / 每个选项页面中央的切换按钮。

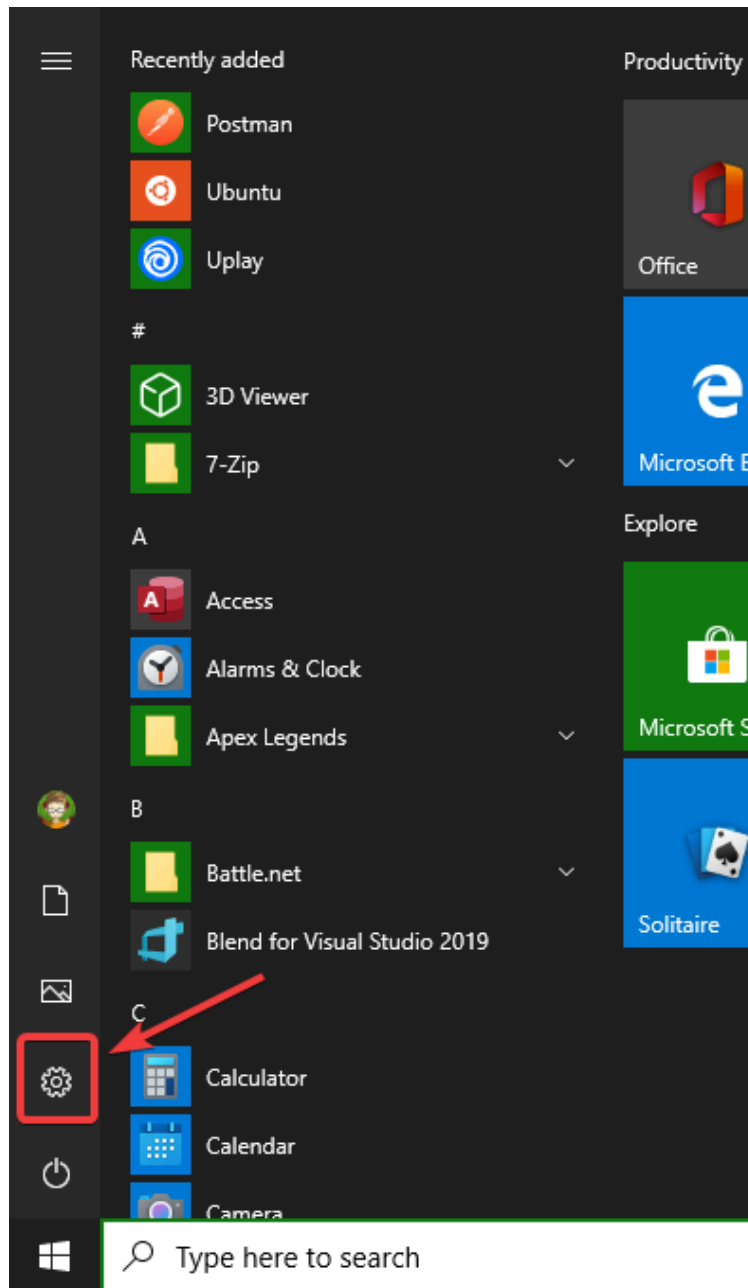
Whitelisting Apps / 将应用加入允许列表

Alternatively, you can add exceptions to the Firewall for any FRC programs you are having issues with.

也可以为遇到问题的 FRC 程序添加防火墙例外规则。

Click **Start** -> **Settings**

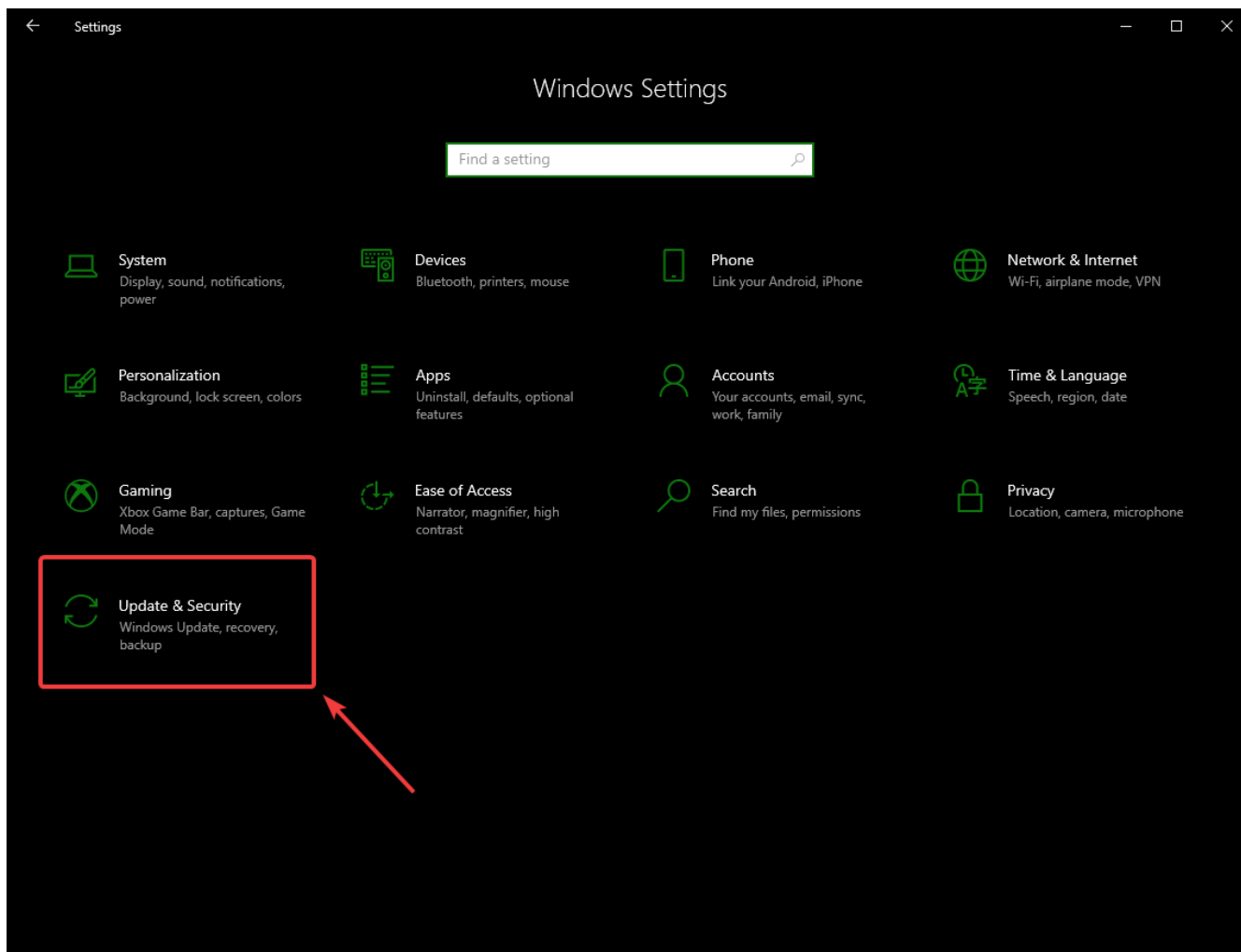
点击**开始** -> **设置**。



The settings gear icon in the start menu. / 开始菜单中的“设置”齿轮图标。

Click **Update & Security**

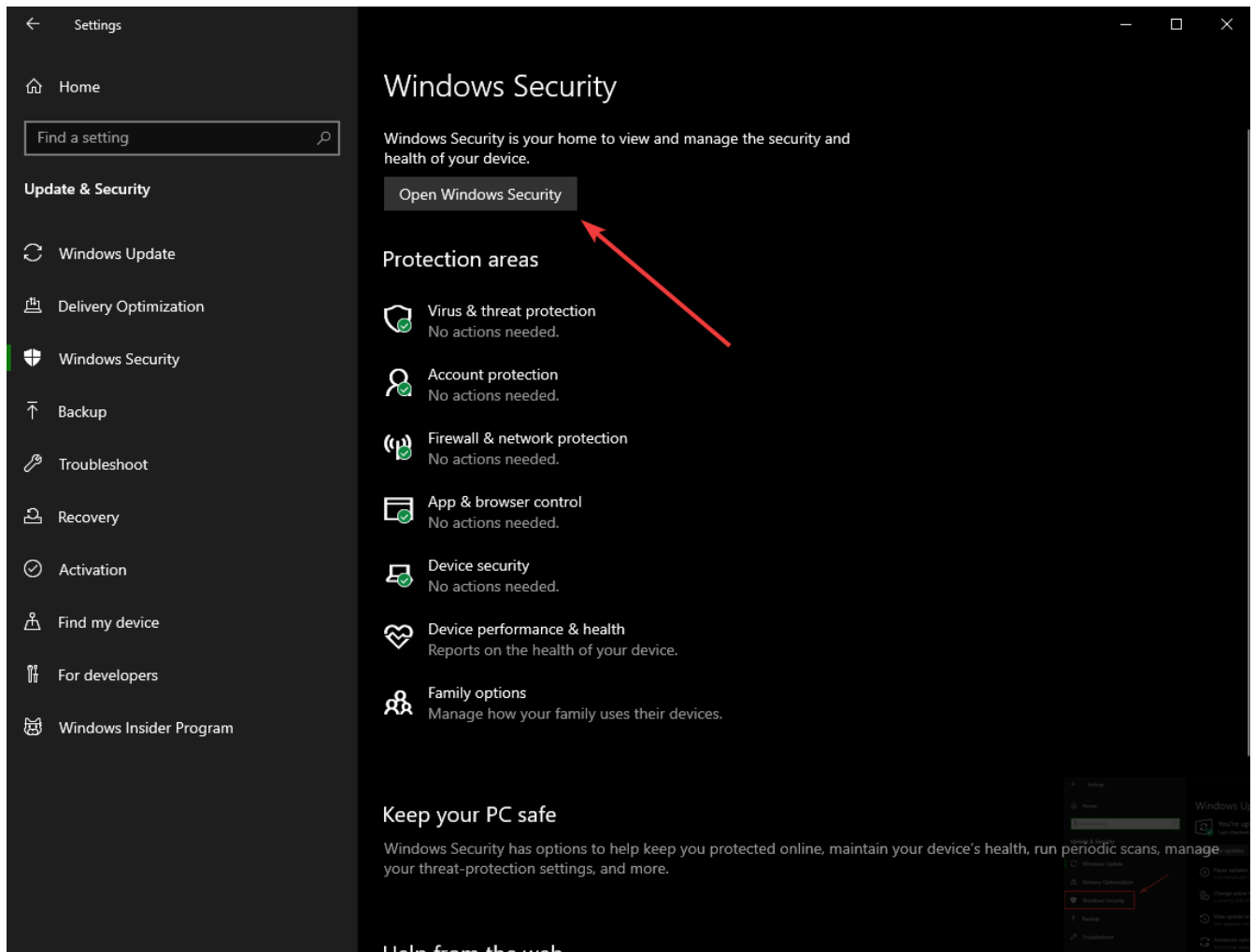
点击**更新和安全**。



Update & Security category. / “更新和安全”类别。

In the right pane, select **Open Windows Security**

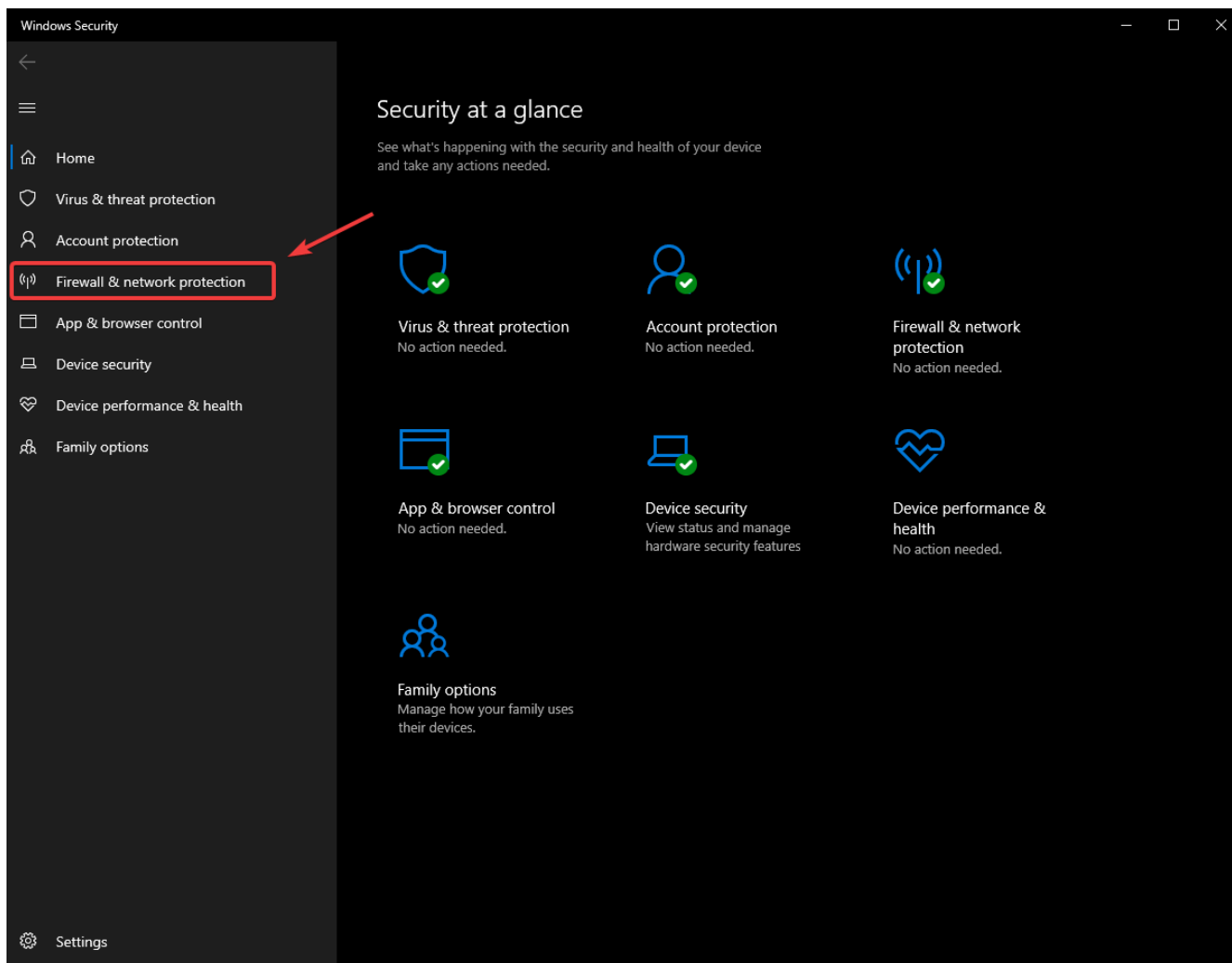
在右侧窗格中，选择**打开 Windows 安全中心**。



Choose "Open Windows Security" / 选择“打开 Windows 安全中心”。

In the left pane, select **Firewall and network protection**

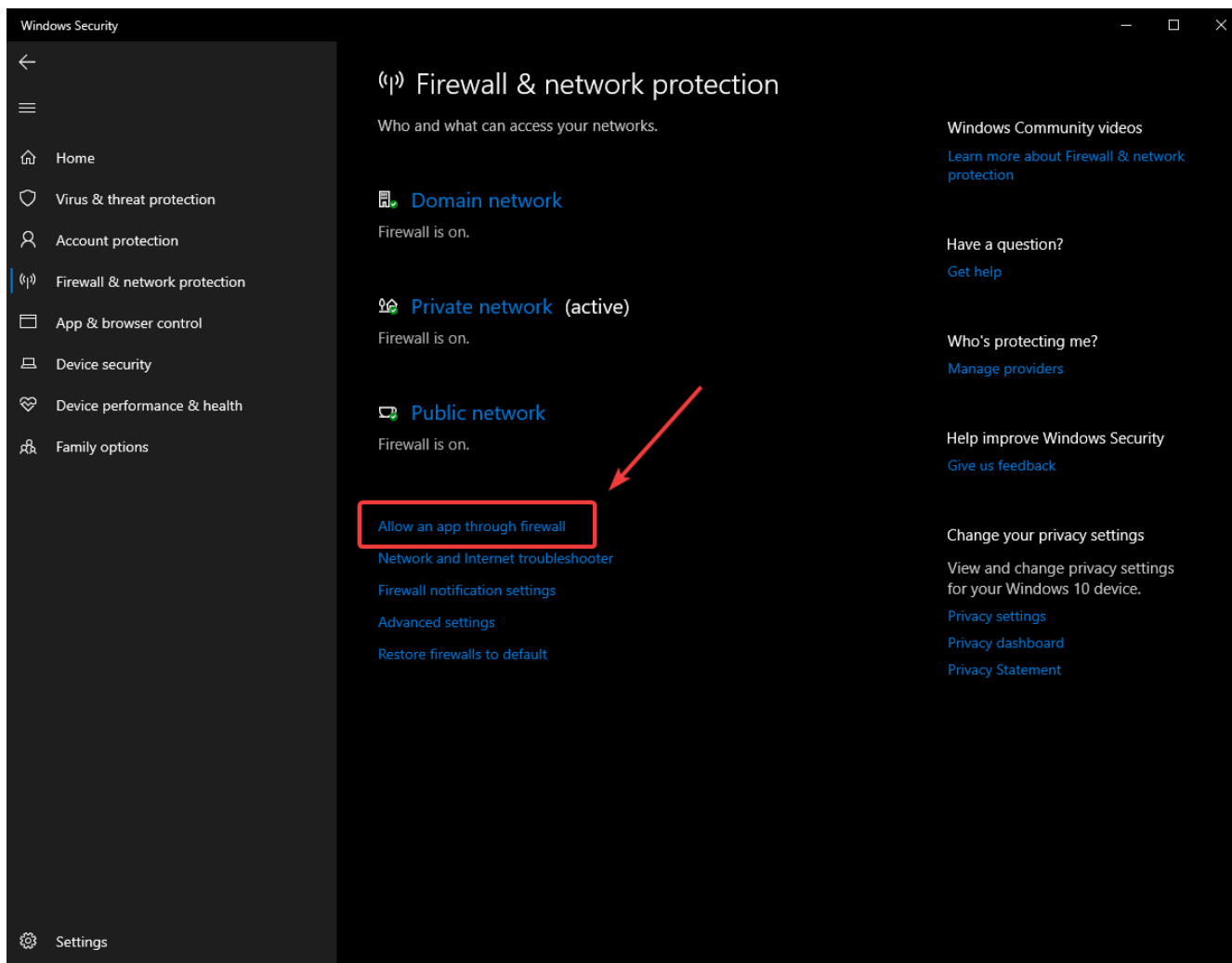
在左侧窗格中，选择**防火墙和网络保护**。



Choose "Firewall & Network protection" in the left pane. / 选择左侧窗格中的“防火墙和网络保护”。

At the bottom of the window, select **Allow an app through firewall**

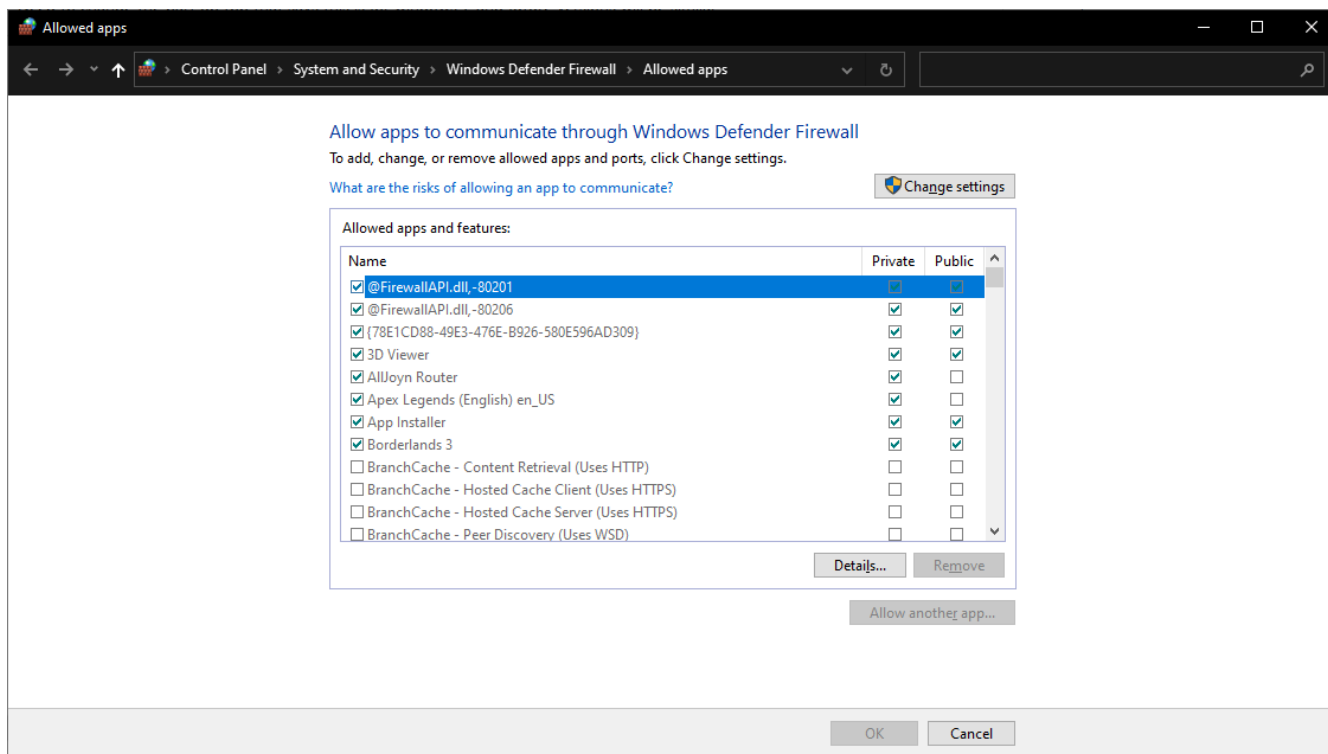
在窗口底部，选择**允许应用通过防火墙**。



Choose "Allow an app through firewall" in the center of the screen. / 选择屏幕中央的“允许应用通过防火墙”。

For each FRC program you are having an issue with, make sure that it appears in the list and that it has a check in each of the 3 columns. If you need to change a setting, you made need to click the **Change settings** button in the top right before changing the settings. If the program is not in the list at all, click the **Allow another program...** button and browse to the location of the program to add it.

对每个出现问题的 FRC 程序，确认它出现在列表中，并且在 3 个列中均已勾选。如果需要更改设置，可能必须先点击右上角的**更改设置**按钮。如果程序完全不在列表中，请点击****允许其他程序...****按钮，浏览并添加该程序所在的位置。



Modify the list presented using the checkboxes to the right of each item. / 使用每项右侧的复选框修改显示的列表。

Measuring Bandwidth Usage / 测量带宽使用量

On the FRC® Field each team is allocated limited network bandwidth (see R704 in the 2026 manual). Some teams may wish to measure their overall bandwidth consumption. This document details how to make that measurement.

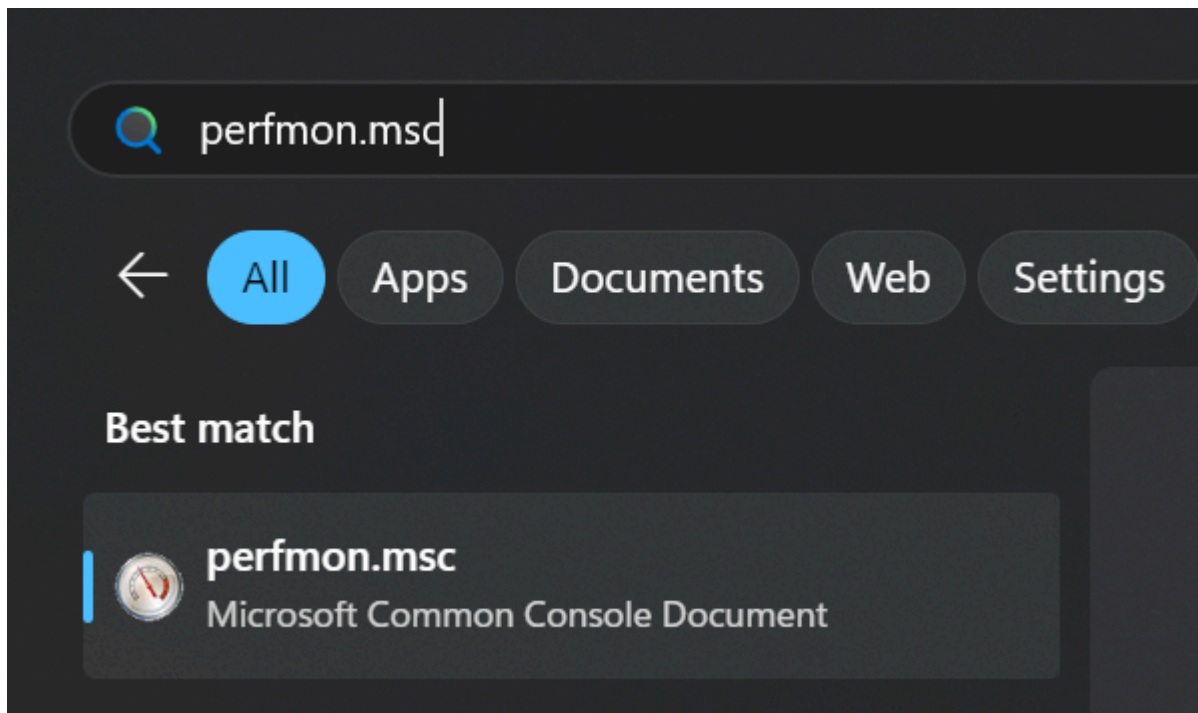
在 FRC® 场地上，每支队伍可用的网络带宽有限（参见 2026 年手册中的 R704）。有些队伍可能希望测量总体带宽消耗。本文详细介绍测量方法。

Measuring Bandwidth Using the Performance Monitor (Win 7/10) / 使用性能监视器测量带宽 (Win 7/10)

Windows contains a built-in tool called the Performance Monitor that can be used to monitor the bandwidth usage over a network interface.

Windows 内置名为“性能监视器”的工具，可用于监测某个网络接口的带宽使用量。

Launching the Performance Monitor / 启动性能监视器

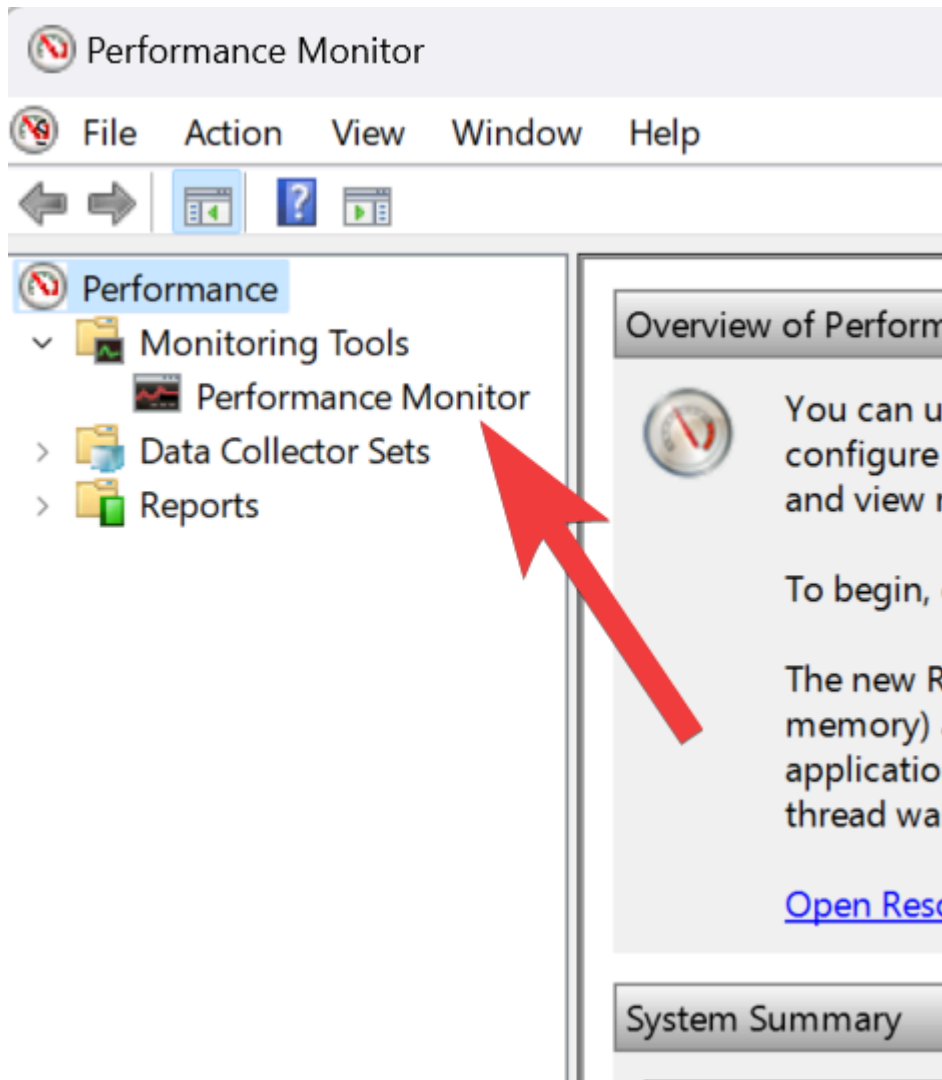


Using the start menu to type "perfmon.msc". / 使用开始菜单输入“perfmon.msc”。

Open the Start menu and in the search box, type `perfmon.msc` and press Enter.

打开“开始”菜单，在搜索框中输入 `perfmon.msc`，然后按 Enter。

Open Real-Time Monitor / 打开实时监视器

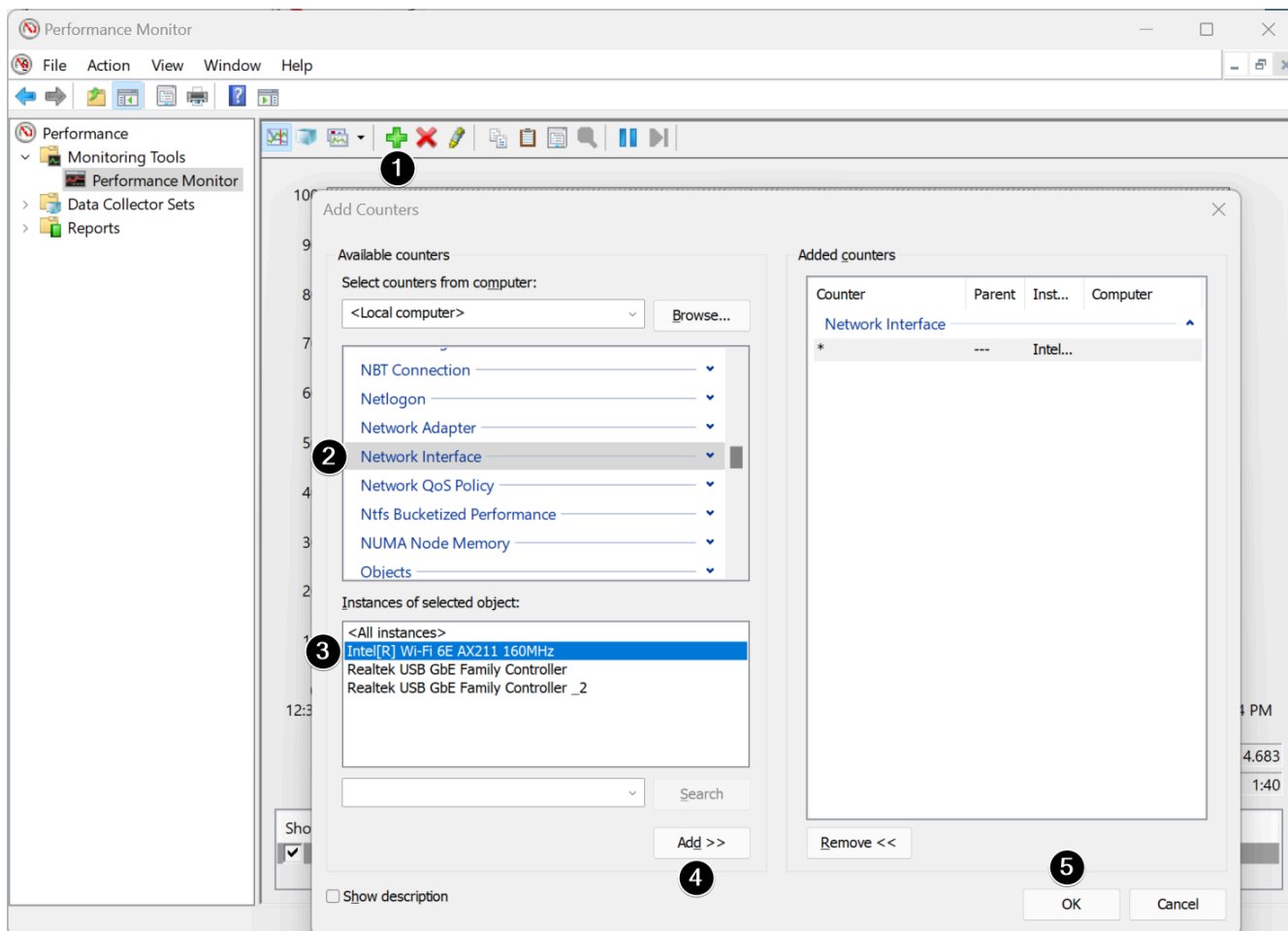


Click "Performance Monitor" under "Monitoring Tools" in the tree view. / 在树形视图的“监视工具”下点击“性能监视器”。

In the left pane, click **Performance Monitor** to display the real-time monitor.

在左侧窗格中点击**性能监视器**，显示实时监视器。

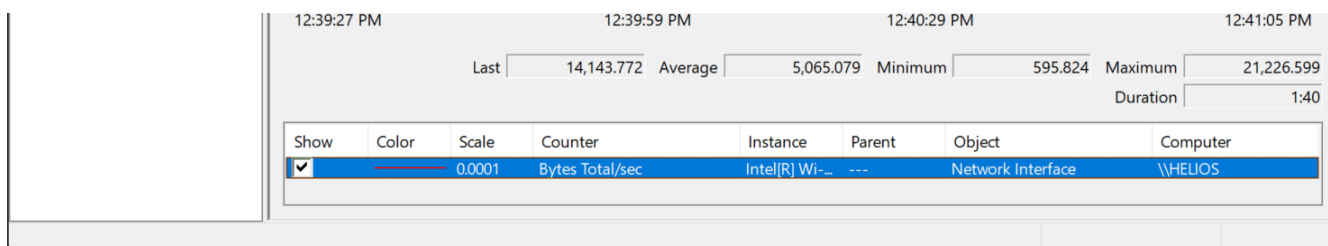
Add Network Counter / 添加网络计数器



The "Add Counters" screen showing the steps below. / “添加计数器”窗口，展示以下步骤。

1. Click the green plus near the top of the screen to add a counter
点击屏幕上方附近的绿色加号，添加一个计数器。
2. In the top left pane, locate and click on **Network Interface** to select it
在左上窗格中找到并点击 **Network Interface**，将其选中。
3. In the bottom left pane, locate the desired network interface (or use All instances to monitor all interfaces)
在左下窗格中找到所需的网络接口（或使用 All instances 监测所有接口）。
4. Click **Add >>** to add the counter to the right pane.
点击**添加 >>**，将计数器添加到右侧窗格。
5. Click **OK** to add the counters to the graph.
点击**确定**，将计数器添加到图表。

Remove Extra Counters / 删除多余的计数器

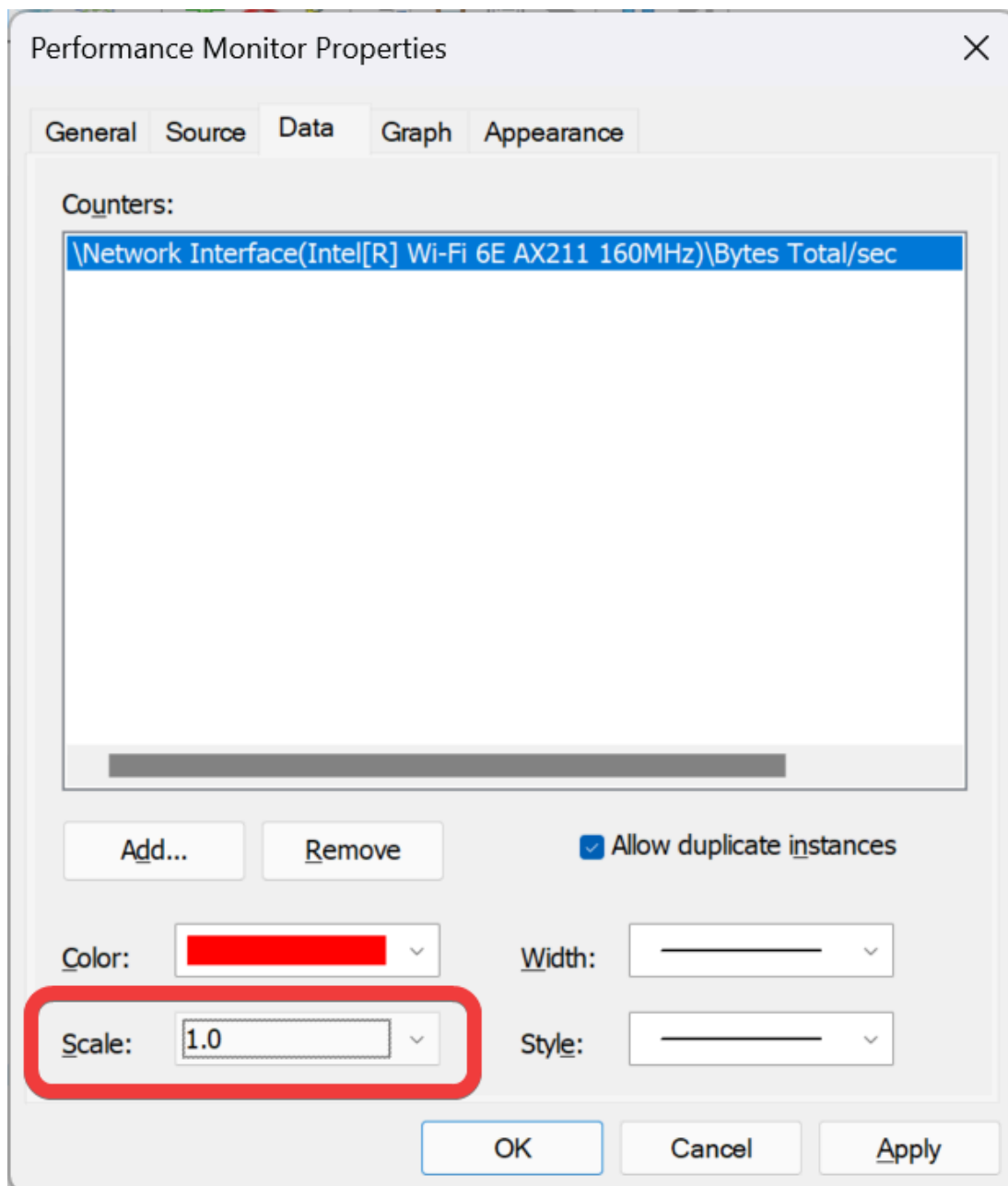


Removing the counter at the bottom of the graph screen. / 删除图表窗口底部的计数器。

In the bottom pane, select each counter other than **Bytes Total/sec** and press the **Delete** key. The **Bytes Total/sec** entry should be the only entry remaining in the pane.

在底部窗格中，选中 **Bytes Total/sec** 以外的各个计数器，并按 **Delete** 键。窗格中应只剩下 **Bytes Total/sec**。

Configure Data Properties / 配置数据属性

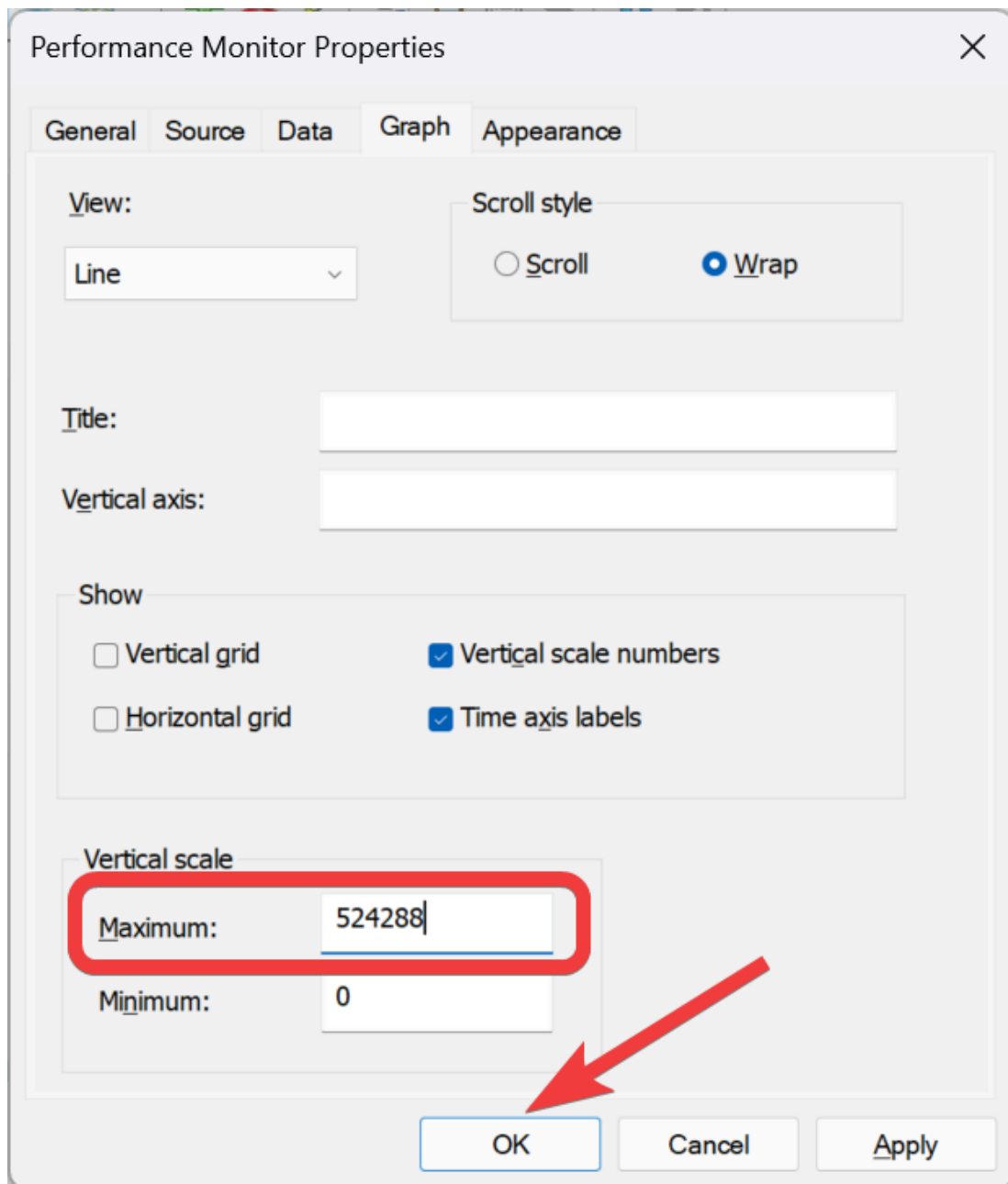


Scale dropdown highlighted on the Performance Monitor Properties page. / 性能监视器属性页面中突出显示了“缩放比例”下拉菜单。

Press **Ctrl+Q** to bring up the Properties window. Click on the dropdown next to **Scale** and select **1.0**. Then click on the **Graph** tab.

按 **Ctrl+Q** 打开“属性”窗口。点击 **Scale** 旁的下拉菜单，选择 **1.0**。然后点击**图形**选项卡。

Configure Graph Properties / 配置图表属性

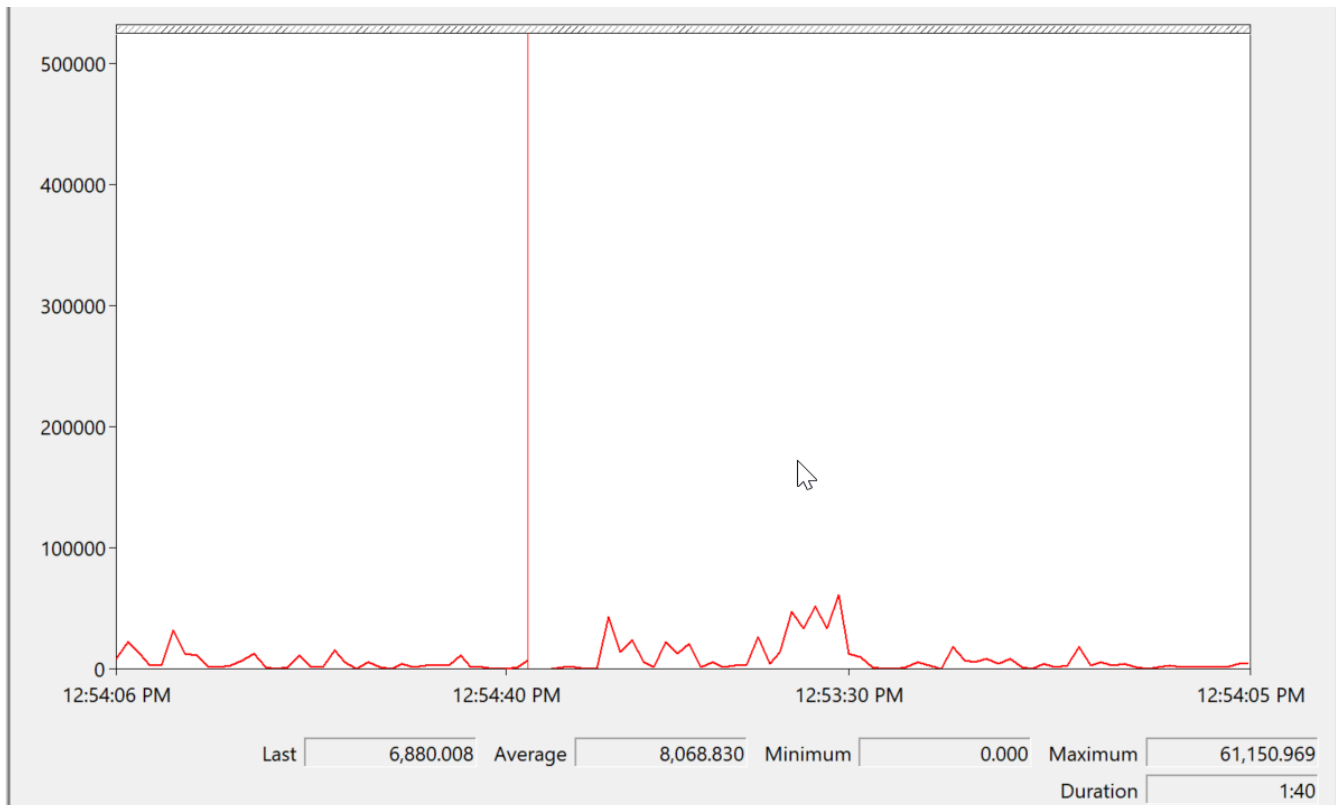


Moving to the Graph tab the Maximum field under Vertical Scale is highlighted. / 切换到“图形”选项卡后，“垂直刻度”下的“最大值”字段被突出显示。

In the **Maximum Box** under **Vertical Scale** enter 524288 (this is 4 Megabits converted to Bytes). If desired, turn on the horizontal grid by checking the box. Then click **OK** to close the dialog.

在 **Vertical Scale** 下的 **Maximum Box** 中输入 524288（即 4 Megabits 换算为 Bytes）。如有需要，可以勾选复选框以显示水平网格。然后点击**确定**关闭对话框。

Viewing Bandwidth Usage / 查看带宽使用量



Observing the bandwidth usage on the chart screen. / 查看图表屏幕上的带宽使用量。

You may now connect to your robot as normal over the selected interface (if you haven't done so already). The graph will show the total bandwidth usage of the connection, with the bandwidth cap at the top of the graph. The Last, Average, Min, and Max values are also displayed at the bottom of the graph. Note that these values are in Bytes/Second meaning the cap is 524,288. With just the Driver Station open you should see a flat line at ~100000 Bytes/Second.

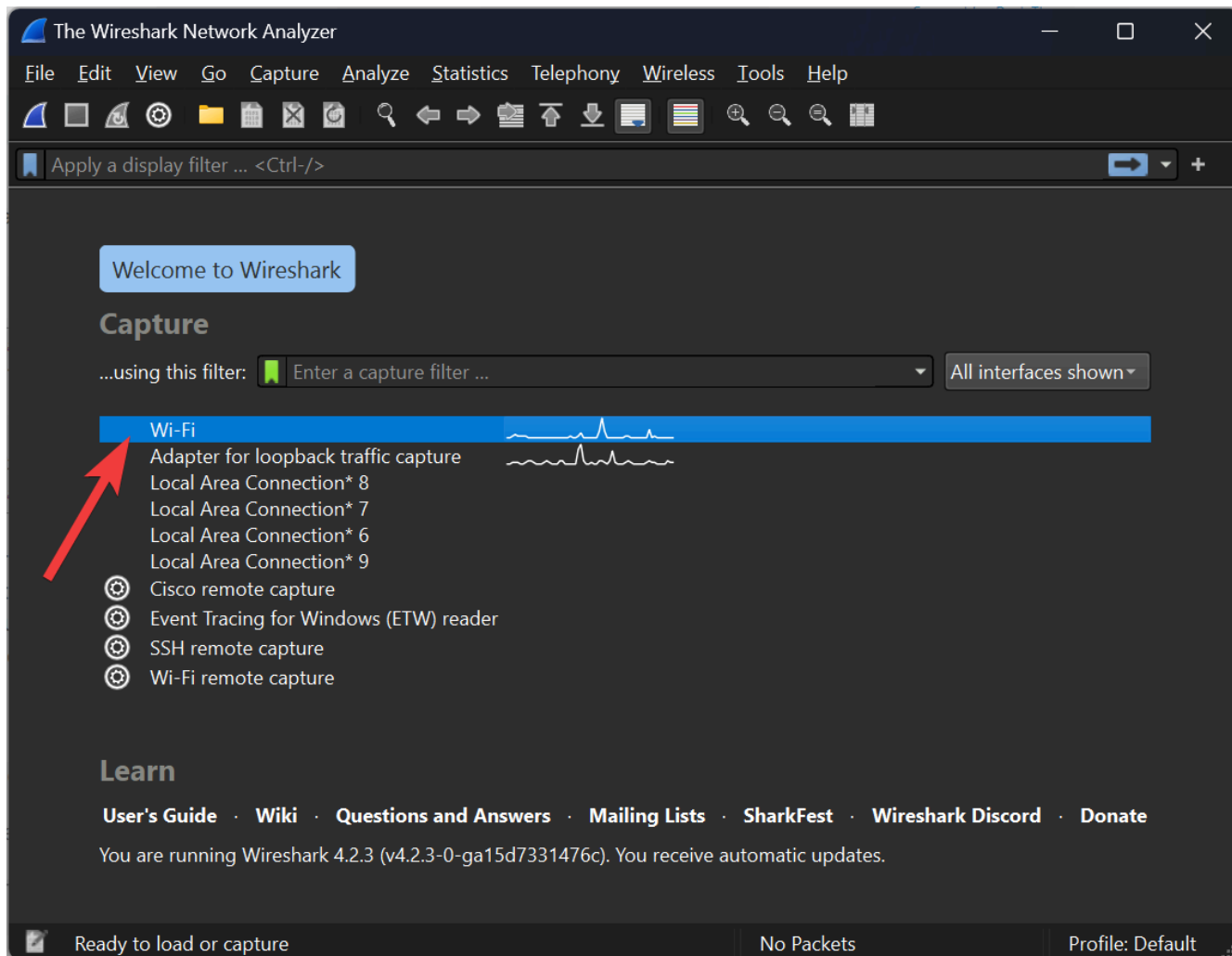
现在可以通过所选接口像平常一样连接机器人（如果尚未连接）。图表会显示连接的总带宽使用量，带宽上限位于图表顶部。图表底部还显示 Last、Average、Min 和 Max 值。注意，这些值的单位为 Bytes/Second，因此上限为 524,288。只打开 Driver Station 时，应看到一条约为 ~100000 Bytes/Second 的平直曲线。

Measuring Bandwidth Usage using Wireshark / 使用 Wireshark 测量带宽使用量

If you cannot use Performance Monitor, you will need to install a 3rd party program to monitor bandwidth usage. One program that can be used for this purpose is Wireshark. Download and install the latest version of Wireshark for your version of Windows from [this page](#). After installation is complete, locate and open Wireshark. Connect your computer to your robot, open the Driver Station and any Dashboard or custom programs you may be using.

如果无法使用性能监视器，就需要安装第三方程序来监测带宽使用量。Wireshark 是可用于此目的的程序之一。请从[此页面](#)下载并安装适用于你的 Windows 版本的最新 Wireshark。安装完成后，找到并打开 Wireshark。将计算机连接到机器人，打开 Driver Station，以及你可能使用的任何 Dashboard 或自定义程序。

Select the interface and Start capture / 选择接口并开始捕获

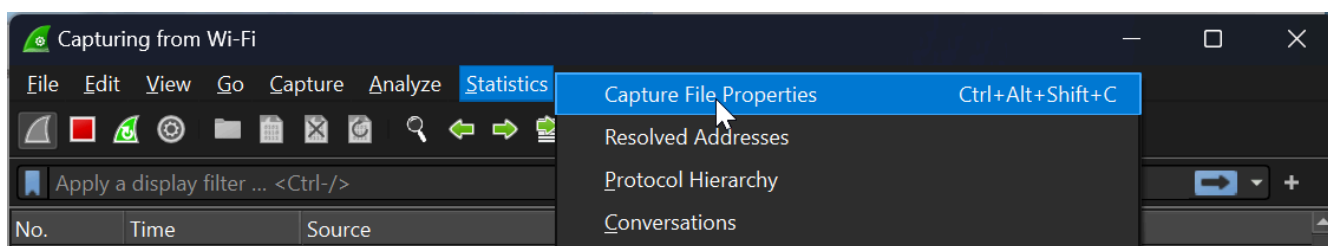


Selecting the Start button and choosing the NIC in Wireshark. / 在 Wireshark 中选择“开始”按钮并选择网卡。

In the Wireshark program on the left side, double click the interface you are using to connect to the robot.

在 Wireshark 程序左侧，双击你用于连接机器人的接口。

Open Capture File Properties / 打开捕获文件属性

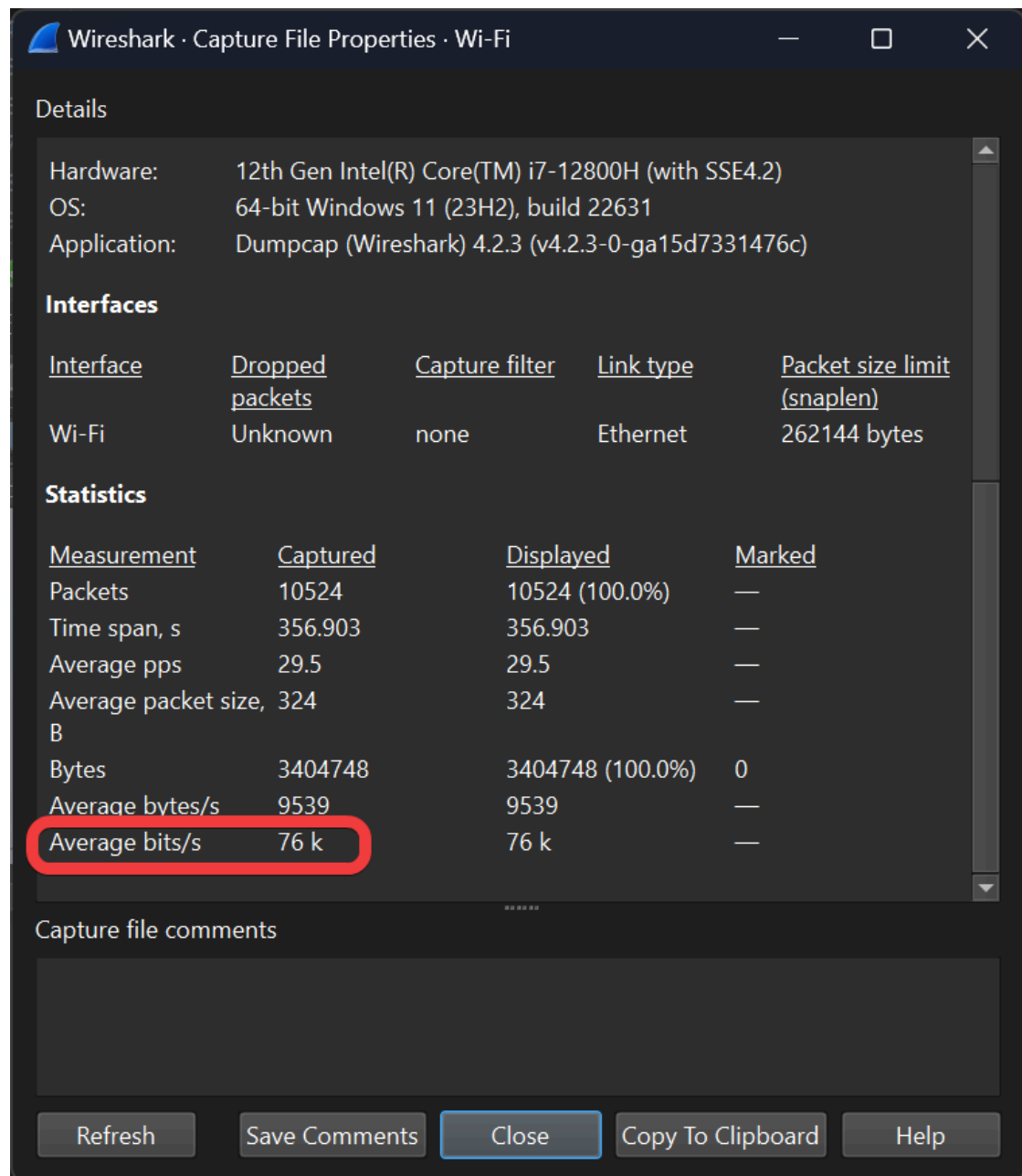


In the menu bar at the top choosing "Statistics" then "Capture File Properties" / 在顶部菜单栏中选择“Statistics”，然后选择“Capture File Properties”。

Let the capture run for at least 1 minute, then click **Statistics** then **Capture File Properties**.

让捕获至少运行 1 分钟，然后点击**Statistics**，再点击**Capture File Properties**。

View Bandwidth Usage / 查看带宽使用量



Looking at the "Average bits/sec" field of the Wireshark summary. / 查看 Wireshark 摘要中的“Average bits/sec”字段。

Average bandwidth usage, in bits/second is displayed near the bottom of the window.

平均带宽使用量以 bits/second 为单位，显示在窗口底部附近。